

CHAPTER THREE

Needs Assessment

Objectives



After reading this chapter, you should be able to

- | | | | |
|-----|---|-----|--|
| 3-1 | Discuss the role of organization analysis, person analysis, and task analysis in needs assessment. | 3-5 | feedback influence performance and learning. |
| 3-2 | Identify different methods used in needs assessment and identify the advantages and disadvantages of each method. | 3-6 | Create conditions to ensure that employees are receptive to training. |
| 3-3 | Discuss the concerns of upper- and mid-level managers and trainers in needs assessment. | 3-7 | Discuss the steps involved in conducting a task analysis. |
| 3-4 | Explain how personal characteristics, input, output, consequences, and | 3-8 | Analyze task analysis data to determine the tasks for which people need to be trained. |
| | | | Explain competency models and the process used to develop them. |

Needs Assessment at NTT and Tata Consultancy Services

Needs assessment is a critical first step in designing new training courses and revising existing ones. Consider how needs assessment is used at NTT and Tata Consultancy Services. NTT is a global technology services company based in London, England. The training and development team at NTT uses a skills matrix to track current and future learning needs for key roles and functions. Employees are asked to rate their capabilities in specific skills on a scale of 1 to 5. Their managers verify the ratings and have the ability to change them after discussions with employees. The training and development team analyzes the data from the matrix by using data dashboards that enable the team and business leaders to compare employees' skill sets across different countries, regions, go-to-market areas, per line manager, and so forth. The team can also present the data through heat maps (a method to visualize data using color in two dimensions), which show areas of strong capability or skills gaps. The information has been very useful for managers, who can quickly see how their teams are currently performing and what their development needs may be in the future. The matrix has also proved a highly effective tool for communicating to leaders the importance of training programs and strategic learning initiatives. Each month the team and senior leaders from each business function review the matrix together and identify areas requiring focus or improvement. For example, one review indicated declining client satisfaction. As a result, the training and development team created training focused on call handling and call flows.

Tata Consultancy Services (TCS) provides information technology and business solutions. TCS wants to build employees' digital competencies. Providing digital solutions to customers are key to the company's continued success. In fact, digital revenues accounted for over 20 percent of TCS's revenue in 2018 and the percentage is expected to increase as customers' needs for digital solutions increase. TCS developed Talent 4.0, a program aimed at developing employees' digital skills. To identify the training best suited for each employee, TCS uses a tool known as T-Factor. The T-Factor measures the depth of employees' specific skill and their ability to apply the skills to certain work areas.

Sources: Based on S. Castellano, "Abandoning Tradition Yet Sticking With Structure," *T+D*, 2020 Best Practices, pp. 31–32; P. Harris, "How to Redefine Excellence," *T+D*, October 2019, pp. 59–61.

Introduction

As discussed in **Chapter One**, "Introduction to Employee Training and Development," effective training practices involve the use of a training design process. The design process begins with a needs assessment. Subsequent steps in the process include ensuring that employees have the motivation and basic skills necessary to learn, creating a positive learning environment, making sure that trainees use learned skills on the job, choosing the training method, and evaluating whether training has achieved the desired outcomes. As the company examples in the chapter opener highlight, before you choose a training method, it is important to determine what type of training is necessary and how it should be delivered. **Needs assessment** refers to the process used to determine whether training is necessary.

Needs assessment typically involves organizational analysis, person analysis, and task analysis.¹ An organizational analysis considers the context in which training will occur. That is, **organizational analysis** involves determining the appropriateness of training, given the company's business strategy, its resources available for training, and support by managers and peers for training activities. You are already familiar with one aspect of organizational analysis. **Chapter Two**, "Strategic Training," discussed the role of the company's business strategy in determining the frequency and type of training.

Person analysis helps identify who needs training. **Person analysis** involves: (1) determining whether performance deficiencies result from a lack of knowledge, skill, or ability (a training issue) or from a motivational or work-design problem; (2) identifying who needs training; and (3) determining employees' readiness for training. **Task analysis** identifies the important tasks and knowledge, skills, and behaviors that need to be emphasized in training for employees to complete their tasks.

Why Is Needs Assessment Necessary?

Needs assessment is important because a manager or other client asking for training—which focuses on closing skill gaps resulting from a lack of knowledge or skill—could really be asking for or need something else, such as a way to motivate employees, change their perspectives or attitudes, or redesign workflow.² If a manager requests training for a performance problem, what he or she is looking for is a solution to a problem that may (or may not) involve training. In conducting a needs assessment, your role is to determine if training is the appropriate solution.

Needs assessment is the first step in the instructional design process, and if it is not properly conducted, any one or more of the following situations may occur:

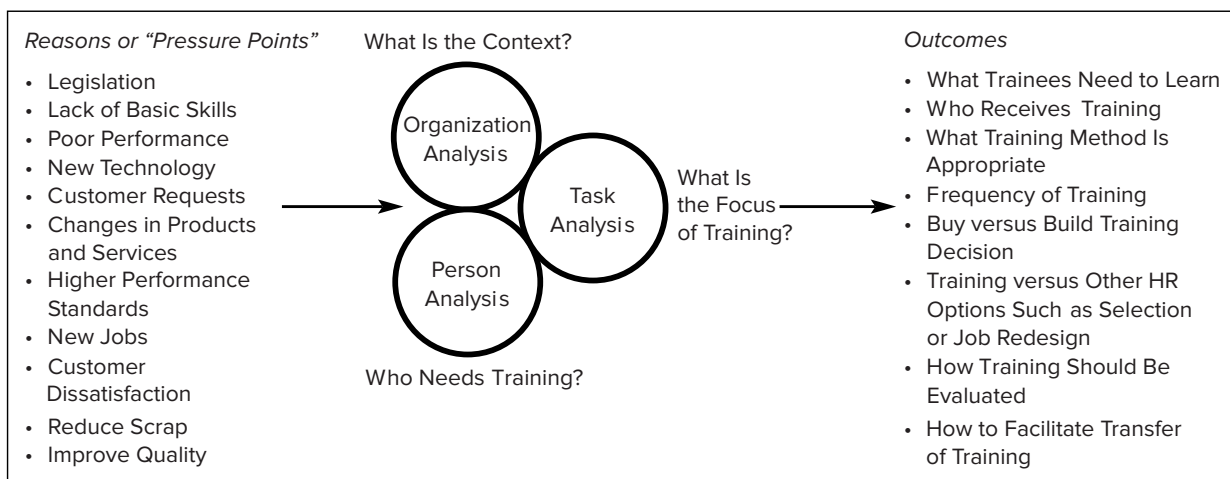
- Training may be incorrectly used as a solution to a performance problem (when the solution should deal with employee motivation, job design, or a better communication of performance expectations).
- Training programs may have the wrong content, objectives, or methods.

- Trainees may be sent to training programs for which they do not have the basic skills, prerequisite skills, or confidence needed to learn.
- Training may not deliver the expected learning, behavior change, or financial results that the company expects.
- Money may be spent on training programs that are unnecessary because they are unrelated to the company's business strategy.

Figure 3.1 shows the three types of analysis involved in needs assessment and the causes and outcomes that result. There are many different causes or “pressure points” that suggest that training is necessary. These pressure points include performance problems, new technology, internal or external customer requests for training, job redesign, new legislation, changes in customer preferences, the introduction of new products or services or changes to existing ones, and employees’ lack of basic skills. For example, consider the pressure points that suggested training was necessary at Accenture and ArcBest.³ Accenture’s cybersecurity training program is based on issues identified by its cyber-intelligence research teams. These issues include trending and emerging security threats and data on security incidents experienced by employees. Simulation scenarios included in the training program require trainees to respond to these issues. ArcBest, a transportation company, introduced new technology to improve driver safety, such as forward-facing cameras, impact sensors, engine monitoring, GPS information, collision detection, incident review, and new electronic logging devices. The implementation and transition to this system involved training over 5,000 drivers and managers.

Note that these pressure points do not automatically mean that training is the correct solution. For example, consider a delivery truck driver whose job is to deliver anesthetic gases to medical facilities. The driver mistakenly hooks up the supply line of a mild anesthetic to the supply line of a hospital’s oxygen system, contaminating the hospital’s oxygen supply. Why did the driver make this mistake, which is clearly a performance problem? The driver may have made this mistake because of a lack of knowledge about the appropriate line hookup for the anesthetic, because of anger over a requested salary increase that his manager recently denied, or because of mislabeled valves for connecting the gas supply. Only the lack of knowledge can be addressed by training. The other pressure points require reviewing and making decisions related to the driver’s anger-motivated behavior (fire the driver) or the design of the work environment (remind supervisors and drivers to check that valves and hookup lines are properly labeled at all work sites).

FIGURE 3.1 Causes and Outcomes of Needs Assessment



What outcomes result from a needs assessment? Needs assessment provides important input into most of the remaining steps in the training design. As shown in **Figure 3.1**, the needs assessment process results in information related to who needs training and what trainees need to learn, including the tasks in which they need to be trained, plus any other knowledge, skill, behavior, or additional job requirements. Needs assessment helps determine whether the company will outsource its training (i.e., purchase training from a vendor or consultant) or develop training through internal resources. Determining exactly what trainees need to learn is critical for the next step in the instructional design process: identifying learning outcomes and objectives. **Chapter Four**, “Learning and Transfer of Training,” explores identifying learning outcomes and learning objectives and creating a training environment so that learning occurs and is used on the job. Through identifying the learning outcomes and resources available for training, the needs assessment also provides information that helps the company choose the appropriate training or development method (discussed in **Part Three** of this book). Needs assessment also provides information regarding the outcomes that should be collected to evaluate training effectiveness. The process of evaluating training is discussed in **Chapter Six**, “Training Evaluation.”

Who Should Participate in Needs Assessment?

Because the goal of needs assessment is to determine whether a training need exists, who it exists for, and for what tasks training is needed, it is important that all stakeholders are included in the needs assessment. **Stakeholders** include persons in the organization who have an interest in training and development and whose support is important for determining its success (or failure). Stakeholders include company leaders and top-level managers, mid-level managers, trainers, and employees who are end users of learning. There are several ways to ensure that stakeholders are involved in needs assessment. One way is through establishing formal advisory groups that meet on a regular basis to discuss learning issues. Another way is to ensure that relevant stakeholders are included in interviews, focus groups, crowdsourcing, and surveys used for needs assessment. Traditionally, only trainers were concerned with the needs assessment process. But, as **Chapter Two** showed, as training is increasingly being used to help the company achieve its strategic goals, both upper- and mid-level managers are becoming involved in the needs assessment process.

Table 3.1 shows the questions that company leaders, mid-level managers, trainers, and employees are interested in answering for organizational analysis, person analysis, and task analysis. Company leaders include directors, CEOs, and vice presidents. Company leaders view the needs assessment process from the broader company perspective rather than focusing on specific jobs. Company leaders are involved in the needs assessment process to identify the role of training in relation to other human resource practices in the company (e.g., selection and compensation of employees). Company leaders want training to anticipate needs and to be aligned with where the business is going. Training and development need to improve employee performance in such a way that they support the business strategy. Learning efforts (training, development, knowledge management) need to take an integrated and holistic approach—rather than consist of a series of fragmented courses or programs—and add value to the company. Company leaders are also involved in identifying what business functions or units need training (person analysis) and in determining if the company has the necessary knowledge, skills, and abilities in its workforce to meet the company’s strategy and be competitive in the marketplace.

Mid-level managers are more concerned with how training may affect the attainment of financial goals for the particular units they supervise. As a result, for mid-level managers, organizational analysis focuses on identifying: (1) how much of their budgets they want to devote to training; (2) the types of employees who should receive training (e.g., engineers, or core employees who are directly involved in producing goods or providing services); and (3) for which jobs training could make a difference in terms of improving products or customer service.

TABLE 3.1 Key Concerns of Company Leaders, Mid-Level Managers, Trainers, and Employees in Needs Assessment

	Company Leaders	Mid-Level Managers	Trainers	Employees
Organizational Analysis	Is training important to achieve our business objectives? How does training support our business strategy?	Do I want to spend money on training? How much?	Do I have the budget to buy training services? Will managers support training?	Is learning rewarded? Can I advance my career? Does my manager encourage and allow me to attend training? Can I access learning?
	What are the threats to our talent base?	How will training and development help meet my business goals? Are we retaining top talent?		
Person Analysis	Do employees in specific functions or business units need training?	Who should be trained? Managers? Professionals? Core employees?	How will I identify which employees need training?	Do I want to learn? Can I learn the training content? Will the content be valuable to me?
	What do employees need to do to accomplish our business objectives?			
Task Analysis	Does the company have people with the knowledge, skills, and abilities or competencies needed to compete in the marketplace?	For what jobs can training make the biggest difference in product quality or customer service?	For what tasks should employees be trained? What knowledge, skills, ability, or other characteristics are necessary?	What knowledge, skills, tasks, or competencies do I need for my current job? Future jobs? My career?

A **learning council** is a group of key stakeholders, typically including executives, company leaders, and other important stakeholders who are asked to provide insight and recommendations about the company's training needs. Consider how Vi and AARP use learning councils in needs assessment.⁴ For example, Vi is a Chicago-based company that operates 10 luxury senior living facilities across the United States. Vi's learning team relies on a learning council consisting of executive leaders to annually help identify and prioritize train-

ing needs. The learning council considers training needs based on business problems, resident care, service quality, safety, financial results, sales, retention, and employee attitude survey results. Managers are asked to provide their reactions to the needs identified by the learning council. Based on this input, the learning team develops their learning strategy. AARP, a non profit organization serving individuals 50 years and older, relied on its learning council to develop its learning strategy goals. The learning council included key stakeholders from each of the organization's business units. The learning council is charged with evaluating requests for new training programs and reviewing existing courses.

As discussed in **Chapter Two**, trainers (including training managers and instructional designers) need to consider whether training is aligned with the business strategy. However, trainers are primarily interested in needs assessment to provide them with the information they need to administer, develop, and support training programs. This information includes determining if training should be purchased or developed in-house, identifying the tasks for which employees need to be trained, and determining upper- and mid-level managers' interest in and support for training.

Employees have several interests in needs assessment. From an organizational perspective, they are concerned with how the company values learning: Is learning rewarded? Does learning help them improve their job performance or meet their career goals? Is it easy to get access to formal and informal learning opportunities? They also want to know if their manager can be expected to encourage them to take courses and programs or informally learn, and whether they will be given support in applying what they have learned. Employees have to determine whether they are motivated to learn, as well as what tasks, knowledge, skills, or competencies they need for their current job or career.

Company leaders are usually involved in determining whether training meets the company's strategy and then providing the appropriate financial resources. Upper-level managers are not usually involved in identifying which employees need training, the tasks for which training is needed, or the knowledge, skills, abilities, and other characteristics needed to complete those tasks. This is the role of subject-matter experts. **Subject-matter experts (SMEs)** are employees, academics, managers, technical experts, trainers, and even customers or suppliers who are knowledgeable with regard to (1) training issues, including tasks to be performed; (2) the knowledge, skills, and abilities required for successful task performance; (3) the necessary equipment; and (4) the conditions under which the tasks have to be performed. A key issue with SMEs is making sure that they are knowledgeable about the content that training must cover, as well as realistic enough to be able to prioritize what content is critical to cover in the time allotted for the subject in the training curriculum. SMEs must also have information that is relevant to the company's business and have an understanding of the company's language, tools, and products. There is no rule regarding how many types of employees should be represented in the group conducting the needs assessment. Still, it is important to get a sample of **job incumbents** (employees who are currently performing the job) involved in the process because they tend to be most knowledgeable about the job. Also, these employees can be a great hindrance to the training process if they do not feel they have had input into the needs assessment.

MasTec, a construction company that engineers, procures, constructs, and maintains the infrastructures for electric power transmission and distribution, oil and natural gas pipelines, and communications companies, wanted to develop an online learning management system through which employees could access training and development courses.⁵ MasTec conducted a needs assessment to determine the technology and functionality that was needed to support new training programs and to identify unique employee needs. The development team started by conducting a stakeholder analysis. This involved considering who would be involved in the process, understanding how to partner with them, and determining what type of information they could offer. It included meeting with safety team leaders, trainers, and construction crew members, observing employees performing their jobs, and attending existing training classes. The development team recorded every need and request made throughout this process. As a result of this analysis, it identified four goals for the learning management system. These goals included (1) increasing the accessibility of training content; (2) increasing

the flexibility and variety in how training is delivered and completed; (3) improving the training registration process for employees; and (4) creating reporting tools to make training requirements, participation, and completion visible to employees, their managers, and the employee development group.

At Fatima Fertilizer Company in Pakistan, engineers or technicians who have become training coordinators work with employees and managers to measure performance and determine training needs based on the knowledge and competency goals established for each department. Each department submits monthly training reports to the training and development team. Executive and department committees meet quarterly with the training and development team to review their progress toward meeting the knowledge and competency goals and to make adjustments in training plans.⁶

Methods Used in Needs Assessment

Several methods are used to conduct needs assessment, including observing employees performing the job; interviewing SMEs; asking SMEs to complete surveys designed to identify the tasks and knowledge, skills, abilities, and other characteristics required for a job; conducting focus groups with SMEs; reading technical manuals and other documentation; using technology; and using historical data. **Table 3.2** presents the advantages and disadvantages of each method.

TABLE 3.2 Advantages and Disadvantages of Needs Assessment Techniques

Technique	Advantages	Disadvantages
Observation	- Generates data relevant to work environment - Minimizes interruption of work	- Needs skilled observers - Employees' behavior may be affected by being observed
Surveys	- Inexpensive - Can collect data from a large number of persons - Data easily summarized	- Requires time - Possible low return rates; inappropriate responses - Lacks detail - Only provides information directly related to questions asked
Interviews	- Good at uncovering details of training needs, as well as causes of and solutions to problems - Can explore unanticipated issues that come up - Questions can be modified	- Time consuming - Difficult to analyze - Needs skilled interviewers - Can be threatening to SMEs - Difficult to schedule - SMEs provide only information they think you want to hear
Focus groups, Crowdsourcing	- Useful with complex or controversial issues that one person may be unable or unwilling to explore - Questions can be modified to explore unanticipated issues - Reduces risk that training based on needs assessment will be rejected by stakeholders	- Time-consuming to organize - Group members provide only information they think you want to hear

Technique	Advantages	Disadvantages
Documentation (technical manuals, records, research studies)	• Good source of information on procedure • Objective • Good source of task information for new jobs and jobs in the process of being created	• You may be unable to understand technical language • Materials may be obsolete
Technology	• Objective • Minimizes interruption of work • Requires limited human involvement • Data can be quickly summarized into reports	• May threaten employees • Manager may use information to punish rather than train
Historical data reviews	• Provide data related to performance and practices	• Available data may be inaccurate or incomplete, or may not fully represent performance

Sources: Based on A. Kuzel, "How to Conduct a Learning Audit," *Chief Learning Officer*, November/December 2016, pp. 23–25, 66; S. V. Steadham, "Learning to Select a Needs Assessment Strategy," *Training and Development Journal*, January 1980, pp. 56–61; R. J. Mirabile, "Everything You Wanted to Know About Competency Modeling," *Training and Development*, August 1997, p. 74; K. Gupta, *A Practical Guide to Needs Assessment* (San Francisco: Jossey-Bass, 1999); M. Casey and D. Doverspike, "Training Needs Analysis and Evaluation for New Technologies Through the Use of Problem-Based Inquiry," *Performance Improvement Quarterly* 18, 1 (2005), pp. 110–24.

Face-to-face and telephone interviews are time-consuming, but more detailed information regarding training needs can be collected. Nestlé USA, a consumer package goods company, identified the need to develop strong leaders, especially employees taking a managerial position for the first time in their careers.⁷ To identify the concerns of employees taking a managerial role, the learning and development team conducted interviews with the new leaders and their managers. The interviews revealed that new leaders struggled with the transition from doing the work alone to delegating and managing others' work. They felt isolated and unprepared for their new position. Their managers found it difficult to identify the skills the new managers needed to develop. From the interviews the learning and development team determined that the new managers needed to build skills in coaching, leading difficult conversations, delegating, leading career conversations, building and managing resiliency, and understanding expectations and resources.

The advantage of surveys is that information can be collected from a large number of persons. Also, surveys allow many employees to participate in the needs assessment process. However, when using surveys, it is difficult to collect detailed information regarding training needs. **Focus groups** are a type of SME interview that involves a face-to-face meeting with groups of stakeholders or SMEs in which the questions that are asked relate to specific training needs. For example, the learning team at Cartus uses focus groups consisting of managers and employees from the department for which training is being developed.⁸ The focus groups discuss the department's business goals and the gaps they feel need to be closed to reach those goals. The focus groups help identify and prioritize training needs. Crowdsourcing can also be used for needs assessment. In this context, **crowdsourcing** refers to asking stakeholders to provide information for needs assessment. Computer Services Corporation uses "Ideation," a web-based tool for collaboration and crowdsourcing, to help identify training needs.⁹ The process requires a review team to filter, sort, and build on the best ideas. The process allows the learning department to get a larger number of employees involved in the needs assessment process rather than relying only on interviews with SMEs. It is important to verify the results of interviews,

surveys, and focus groups however, because what employees and managers say they do and what they really do may differ.

For newly created jobs, trainers often do not have job incumbents to rely on for needs assessment information. Technical diagrams, research studies, simulations, and equipment designers can provide information regarding the training requirements, tasks, and conditions under which a job is performed. This information is especially useful for newly created jobs where trainers do not have job incumbents they can observe, survey, interview, or ask to participate in a focus group. LeadingRE used the results of a research study it purchased from a consulting firm as part of its needs assessment.¹⁰ The research study identified residential real estate customers' interests, desires, emotions, attitudes, and lifestyles. LeadingRE used this information to design training to teach its agents the types of questions to ask their customers to better understand their preferences, create stronger relationships, and enhance sales.

Technology that monitors employee behavior and performance also can be used for needs assessment. Software and wearables such as Google glasses can be used to collect data on employee behavior and performance. For example, employees in the melt department at H&H Castings have a demanding and potentially dangerous job that requires them to pour hot aluminum into casting molds.¹¹ To reduce the risk of accidents and improve new employee training, H&H Castings analyzed the job tasks of its melt department employees and identified the procedures needed to complete them. To identify the tasks and procedures, employees wore eye-tracking glasses as they worked. A computer connected to the eye-tracking glasses recorded what employees were looking at as they worked and provided a detailed and precise view of how employees completed their job tasks. The results showed that the work requires a high level of concentration. The perspective obtained from the eye-tracking glasses was used to create videos that are now used for both training new employees and identifying changes that could be made to make the mold-pouring process safer and more efficient. This information also is useful for identifying training needs and providing employees with feedback regarding their skill strengths and weaknesses. In call centers, technology provides an ongoing assessment of performance.¹² An employee who triggers the online system by failing to meet a defined standard, such as receiving more than five callbacks on an unresolved issue, is automatically referred to the appropriate job aid or training event. As shown in **Table 3.2**, online technology offers several advantages: it provides an objective report of behaviors, the data can be quickly summarized into reports, it does not require a trainer or SME to observe or interview employees, and it minimizes work interruptions. However, for technology to be effective, managers need to ensure that the information is used to train employees, rather than punish them. Otherwise, employees will feel threatened, which will contribute to employee dissatisfaction and turnover.

Data reviews can also provide helpful information to determine training needs. Data reviews involve collecting and analyzing performance data from electronic or paper records. It provides information regarding current performance levels, which is useful for identifying gaps between actual and desired performance. For companies that have introduced a new technology, another source of historical data is the help desk that companies often set up to deal with calls regarding problems, deficiencies in training, or deficiencies in documentation, software, or systems.¹³ Help-desk management software can categorize and track calls and questions by application, by caller, or by vendor. Report creation capability built into the software makes it easy to generate documents on user problems and identify themes among calls. Analyzing these calls is practical for identifying gaps in training. Common types of call problems can be analyzed to determine if they are due to inadequate coverage in the training program and/or inadequate written documentation and job aids used by trainees.

Consider how KCB Bank used data reviews to help identify training needs.¹⁴ In the last three years, the number of in-person transactions made at KCB Group bank branches decreased from 36 to 12 percent as more customers relied on online banking tools. To determine if its relationship managers needed training to better serve customers due to the increase in digital banking, the talent development team conducted a needs assessment. During the needs assessment, the team considered key success metrics for the branches

including customer wait times, time to serve customers, portfolio of products held per customer, profitability, complaints, and staff productivity. It also visited branches to observe customer interactions and talk to staff about their daily activities and learning gaps.

Because no single method of conducting needs assessment is superior to the others, multiple methods are usually used. The methods vary in the type of information, as well as the level of detail provided. Samsung Electronics uses multiple methods and involves multiple stakeholders to determine training needs.¹⁵ Twice a year at company headquarters, the talent development team holds a business strategy that includes the top 100 leaders from the company's locations from around the world. The purpose of the seminar is to review short- and long-term business plans and goals and identify how training can support them. Also, annually the talent development team conducts employee and manager surveys in each country where it has operations and follows up with interviews to better understand the results. The talent team meets with company leaders from each country to discuss how training could help meet the needs of their business. Based on the information gathered from the surveys, interviews, and meetings, the talent development team prioritizes training needs, how they plan to meet those needs, and the budget needed. Every month the team meets to discuss progress and results, review training and business metrics, and make any adjustments needed to training plans and priorities.

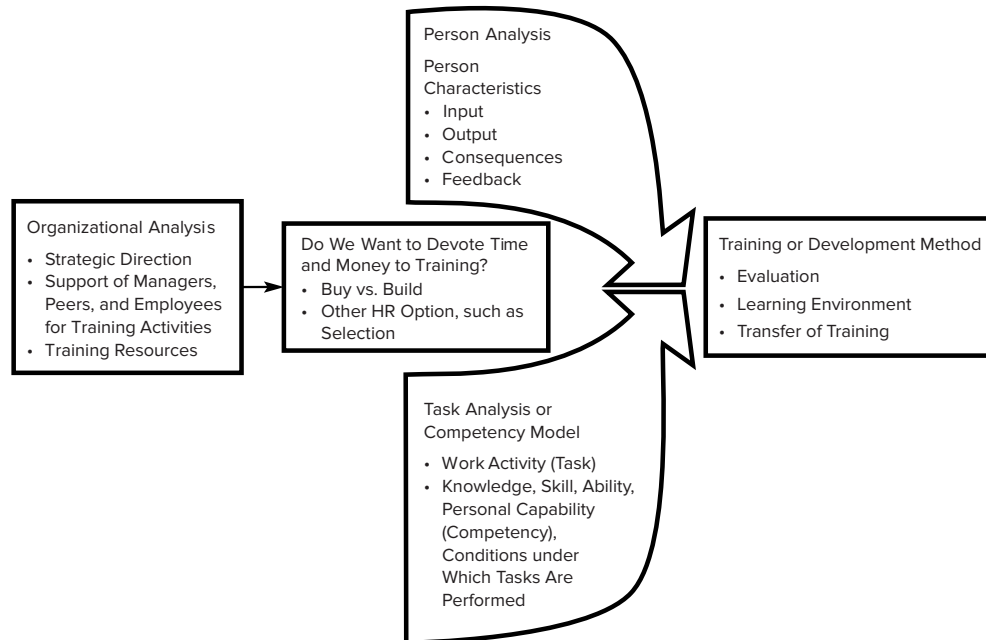
Many companies are also using information about other companies' training practices (a process known as **benchmarking**) to help determine the appropriate type, level, and frequency of training.¹⁶ For example, UPS, Walmart, Comcast, Delta Airlines, Pfizer, and 50 other companies are members of the ATD (Association for Talent Development) forum.¹⁷ Each company completes a common survey instrument, responding to questions on training costs, staff size, administration, design, program development, and delivery. The information is then summarized and shared with the participating companies.

The Needs Assessment Process

This section examines the three elements of needs assessment: organizational analysis, person analysis, and task analysis. **Figure 3.2** illustrates the needs assessment process. While any one analysis can indicate the need for training, companies need to consider the information from all three types of analysis before the decision is made to devote time and money to training. In practice, organizational analysis, person analysis, and task analysis are not conducted in any order. However, because organizational analysis is concerned with identifying whether training suits the company's strategic objectives and whether the company has the budget, time, and expertise for training (the context for training), it is usually conducted first. Person analysis and task analysis are often conducted at the same time because it is difficult to determine whether performance deficiencies are a training problem without first understanding the tasks and the work environment. An initial organizational analysis may suggest that a company does not want to spend financial resources on training. However, if person analysis reveals that a large number of employees lack a skill in an important area that is related to the company's business objectives (such as customer service), upper-level managers may decide to reallocate financial resources for training.

Organizational Analysis

Organizational analysis involves identifying whether training supports the company's strategic direction; whether managers, peers, and employees support training activity; and what training resources are available. **Table 3.3** provides questions that trainers should ask in an organizational analysis. Some combination of documentation, interviews, focus groups, or surveys of managers, individuals in the training function, and employees should be used to answer these questions.

FIGURE 3.2 The Needs Assessment Process**TABLE 3.3** Questions to Ask in an Organizational Analysis

- How might the training content affect our employees' relationship with our customers?
- What might suppliers, customers, or partners need to know about the training program?
- How does this program align with the strategic needs of the business?
- What business and performance results do clients of training and development (typically business leaders) want?
- Should organizational resources be devoted to this program?
- What do we need from managers and peers for this training to succeed?
- What features of the work environment might interfere with training (e.g., lack of equipment, no time to use new skills)?
- Do we have experts who can help us develop the program content and ensure that we understand the needs of the business as we develop the program?
- Will employees perceive the training program as an opportunity? Reward? Punishment? Waste of time?
- Which persons or groups (e.g., employees, managers, vendors, suppliers, and/or program developers) have an interest in seeing training succeed? Whose support do we need?

Sources: Based on D. Robinson, "Transitioning from Order-Taker to Impact-Maker," *T+D*, January 2018, pp. 42–46; F. Nickols, "Why a Stakeholder Approach to Evaluating Training?" *Advances in Developing Human Resources*, February 2005, pp. 121–134; S. Tannenbaum, "A Strategic View of Organizational Training and Learning," in *Creating, Implementing, and Managing Effective Training and Development*, ed. K. Kraiger (San Francisco: Jossey-Bass, 2002), pp. 10–52.

The Company's Strategic Direction

How the company's business strategy influences training was discussed in **Chapter Two**. The strategic role of training influences the frequency and type of training and how the training function is organized in the company. In companies in which training is expected to contribute to the achievement of business strategies and

goals, the amount of money allocated to training and the frequency of training will likely be higher than in companies in which training is done haphazardly or with no strategic intent. For example, companies that believe learning contributes to their competitive advantage or that have adopted high-performance work systems (e.g., teams) are likely to have greater training budgets and conduct more training. The business strategy also influences the type of training. For example, as noted in **Chapter Two**, companies that have adopted a disinvestment strategy are more likely to focus on outplacement assistance and job search skills training than are companies with other strategic initiatives. Last, the greater the strategic role of training, the more likely that the company will organize the training function using the business-embedded or corporate university models. Both of these models emphasize the use of training to help solve business problems.

For example, to stay competitive, IBM has to stay up-to-date on the newest technology and business trends.¹⁸ IBM has to constantly reinvent itself to ensure that it can meet the needs of its customers. This means that employees also must continue to develop new knowledge and skills. From a learning perspective, IBM has to ensure that the learning content it offers, including both face-to-face and online courses, provides employees with the latest knowledge and skills. To accomplish this, IBM keeps track of both how often employees use the learning content it offers as well as how useful that content is, based on employees' evaluations. As a result of reviewing learning content use, IBM eliminated 39 percent of the learning content that few employees found useful or used. This included 7,600 courses!

Support of Managers, Peers, and Employees for Training Activities

A number of studies have found that peer and manager support for training is critical, along with employee enthusiasm and motivation to attend training. The key factors for success are a positive attitude among peers, managers, and employees about participation in training activities; managers' and peers' willingness to provide information to trainees about how they can use the knowledge, skills, or behaviors learned in training to perform their jobs more effectively; and opportunities for trainees to use training content in their jobs.¹⁹ If peers' and managers' attitudes and behaviors are not supportive, employees are not likely to apply training content to their jobs.

Training Resources

It is necessary to identify whether the company has the budget, time, and expertise for training. One of the questions that the company must answer is whether it has the resources (i.e., time, money, and expertise) to build or develop training programs itself or whether it should buy them from a vendor or consulting firm. This is known as the "buy versus build" decision. For example, if the company is installing computer-based manufacturing equipment in one of its plants, it has three possible strategies for dealing with the need to have computer-literate employees. First, the company may decide that, given its staff expertise and budget, it can use internal consultants to train all affected employees. Second, the company may decide that it is more cost effective to identify employees who are computer literate by using tests and work samples. Employees who fail the test or perform below standards on the work sample can be reassigned to other jobs. Choosing this strategy suggests that the company has decided to devote resources to selection and placement rather than training. Third, because it lacks time or expertise, the company may decide to buy training from a consultant. We will discuss how to identify and choose a high-quality vendor or consultant to provide training services in **Chapter Five**, "Program Design."

One way to identify training resources is for companies with similar operations or departments located across the country or the world to share ideas and practices. Kaiser Permanente, a California-based health-care company, is organized by regional business units.²⁰ In each region the company integrates health-care delivery, including hospitals, outpatient services, and insurance providers. One of its concerns was how to provide

consistent and high-quality learning and development opportunities for employees in all of the regions. To do so, the vice president of learning and development created a group, the National Learning Leaders, which includes leaders from account management, sales, compliance, quality improvement, and patient safety. The National Learning Leaders meet three times every year and in smaller working groups in the months when the full membership is not scheduled to meet. They discuss how to develop learning solutions that can be implemented across the business and what kinds of services should be provided by the learning organizations within each region. This has resulted in learning initiatives that are consistent across the regions and delivered in a way that enhances employee participation. For example, based on input from the National Learning Leaders, online and hybrid training courses on patient safety were developed. So far, 19,000 employees have completed these modules, compared to only 50 who attended classroom-based training.

Person Analysis

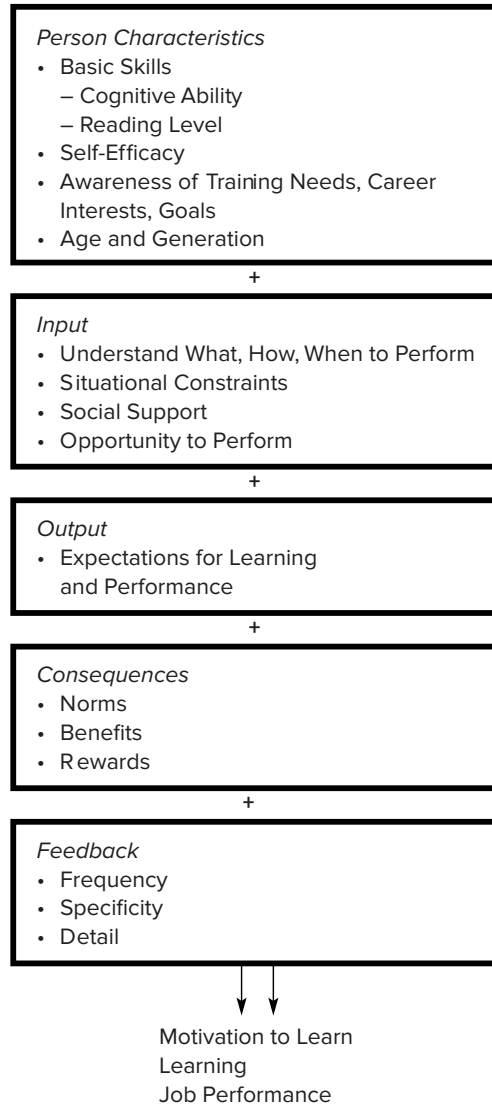
Person analysis helps identify employees who need training, perhaps due to lack of training or poor previous training. This is often referred to as a gap analysis. A **gap analysis** includes determining what is responsible for the difference between employees' current and expected performance. The need for training may result from the pressure points in **Figure 3.1**, including performance problems, changes in the job, or the use of new technology. Person analysis also helps to determine employees' readiness for training. **Readiness for training** refers to whether (1) employees have the personal characteristics (ability, attitudes, beliefs, and motivation) necessary to learn program content and apply it on the job, and (2) the work environment will facilitate learning and not interfere with performance. This process includes evaluating person characteristics, input, output, consequences, and feedback.²¹

A major pressure point for training is poor or substandard performance. Poor performance is indicated by customer complaints, low performance ratings, or on-the-job incidents such as accidents and unsafe behavior. Another potential indicator of the need for training is if the job changes such that current levels of performance need to be improved or employees must be able to complete new tasks.

The Process for Person Analysis

Figure 3.3 shows the process for analyzing the factors that influence performance and learning. **Person characteristics** refer to employee knowledge, skill, ability, and attitudes. **Input** relates to the instructions that tell employees what, how, and when to perform. Input also refers to the resources that the employees are given to help them perform. These resources may relate to equipment, time, or budget. **Output** refers to the job's performance standards. **Consequences** refer to the type of incentives that employees receive for performing well. **Feedback** refers to the information that employees receive while they are performing.

Interviews or questionnaires can be used to measure person characteristics, input, output, consequences, and feedback. For example, a package delivery company believed that lead drivers were valuable for providing on-the-job training for new employees.²² The company employed 110 lead drivers, and the job involved driving, delivery, and bookkeeping duties. The lead drivers benefited from their training role because coaching and training made their jobs more interesting, and the company benefited because on-the-job training was relatively inexpensive and effective. Lead drivers were often able to quickly spot and correct performance problems with new trainees, and they knew the technical aspects of the delivery job quite well. Although many of the lead drivers were already good trainers and coaches, the company believed that they needed to learn how to coach and train the new drivers. The company used interviews to identify what type of coaching and training skills the lead drivers needed. Interviews were conducted with 14 lead drivers, six supervisors, and two regional vice presidents. The interview for the lead drivers consisted of questions such as the following:

FIGURE 3.3 The Process for Analyzing the Factors That Influence Employee Performance and Learning

Sources: R. Jaenke, "Identify the Real Reasons Behind Performance Gaps," *T+D*, August 2013, pp. 76–77; C. Reinhart, "How to Leap over Barriers to Performance," *Training and Development*, January 2000, pp. 20–24; G. Rummier and K. Morrill, "The Results Chain," *T+D*, February 2005, pp. 27–35.

- What types of situations call for coaching on your part?
- What keeps you from being a good coach on the job?
- How do you encourage or motivate other lead drivers? Do you use incentives or rewards? Do you try other things (compliments, personal attention)?
- What common types of performance problems do new hires have?
- What were the biggest problems you encountered as a new coach and trainer? What mistakes did you make? What lessons have you learned over time?
- Tell me about a successful coaching experience and an unsuccessful coaching experience.

Recurring trends in the interview data were noted and categorized. For example, interview questions on obstacles to coaching related to three themes: lack of time to coach, the physical environment (no privacy), and reluctance to coach peers. These three topics were covered in the coaching course.

Person characteristics, input, output, consequences, and feedback influence the motivation to learn. **Motivation to learn** is trainees' desire to learn the content of training programs.²³ Consider how your motivation to learn may be influenced by person characteristics and the environment. You may have no problem understanding and comprehending the contents of this textbook. But your learning may be inhibited because of your attitude toward the course. That is, perhaps you do not believe that the course will be important for your career. Maybe you are taking the course only because it fits your schedule or is required in your degree program. Learning may also be inhibited by the environment. For example, maybe you want to learn, but your study environment prevents you from doing so. Every time you are prepared to read and review your notes and the textbook, your roommates could be having a party. Even if you do not join them, the music may be so loud that you cannot concentrate.

Marriott International, the hotel and restaurant chain, found that personal characteristics were having a significant influence on the success rate of the company's welfare-to-work program.²⁴ This program involves training welfare recipients for jobs in the company's hotels and restaurants. (These types of programs are discussed in greater detail in **Chapter Ten**, "Social Responsibility: Legal Issues, Managing Diversity, and Career Challenges.") Many trainees were unable to complete the training program because of poor attendance due to unreliable child care, drug problems, or abusive partners. As a result, Marriott has instituted strict standards for selecting welfare recipients for the training program. These standards include requiring trainees to have child care, transportation, and housing arrangements. Also, Marriott added an additional drug test during training.

A number of research studies have shown that motivation to learn is related to knowledge gained, behavior changes, or skill acquisition resulting from training.²⁵ Besides considering the factors of person characteristics, input, output, consequences, and feedback in determining whether training is the best solution to a performance problem, managers should also take these factors into account when selecting which employees will attend a training program. These factors relate to the employees' motivation to learn. The following sections describe each of these factors and its relationship to performance and learning.

Person Characteristics

Person characteristics include basic skills, cognitive ability, language skills, and other traits that employees need to perform their jobs or learn in training and development programs effectively. Person characteristics also include employees' age or generation, which might affect how they prefer to learn. As mentioned in **Chapter One**, recent forecasts of workforce skill levels and survey results suggest that companies are experiencing a skills gap. That is, companies are having difficulty finding employees who have the right knowledge, skills, or abilities to fill open positions or to succeed in training to prepare them for today's jobs.

Basic Skills

Basic skills refer to skills that are necessary for employees to perform on the job and to learn the content of training programs successfully. Basic skills include cognitive ability and reading and writing skills. For example, one assumption that your professor is making in this course is that you have the necessary reading level to comprehend this textbook and the other course materials such as PowerPoint slides, videos, or readings. If you lack the necessary reading level, you likely will not learn much about training in this course. As **Chapter One** discussed, recent forecasts of skill levels of the U.S. workforce indicate that managers will likely have to work with employees who lack basic skills. A literacy audit can be used to determine employees' basic skill levels. **Table 3.4** shows the activities involved in conducting a literacy audit.

TABLE 3.4 Steps in Performing a Literacy Audit

Step 1:	Observe employees to determine the basic skills that they need to be successful in their job. Note the materials the employee uses on the job, the tasks performed, and the reading, writing, and computations completed.
Step 2:	Collect all materials that are written and read on the job and identify the computations that must be performed to determine the necessary level of basic skill proficiency. Materials include bills, memos, and forms such as inventory lists and requisition sheets.
Step 3:	Interview employees to determine the basic skills that they believe are needed to do the job. Consider the basic skill requirements of the job yourself.
Step 4:	Determine whether employees have the basic skills needed to perform the job successfully. Combine the information gathered by observing and interviewing employees and evaluating materials they use on the job. Write a description of each job in terms of the reading, writing, and computation skills needed to perform the job successfully.
Step 5:	Develop or buy tests that ask questions relating specifically to the employees' job. Ask employees to complete the tests.
Step 6:	Compare test results (from step 5) with the description of the basic skills required for the job (from step 4). If the level of the employees' reading, writing, and computation skills does not match the basic skills required by the job, then a basic skills problem exists.

Sources: U.S. Department of Education; U.S. Department of Labor, *The Bottom Line: Basic Skills in the Workplace* (Washington, DC: 1988), pp. 14–15.

It is important to note that possession of a high school diploma or a college degree is no guarantee that an employee has basic skills. If participants do not have the fundamental reading, writing, and math skills to understand the training, they will not be able to learn, they will not apply their training to the job (a process known as *transfer*, which is discussed in **Chapter Four**), and the company will have wasted money on training that does not work. Trainers need to evaluate the strengths and weaknesses of trainees before designing a training program. The skill weaknesses that are identified can be used to determine prerequisites that trainees need or must acquire before entering a training program. How do trainers identify skills gaps?²⁶ First, trainers collect general information through position-specific training materials and job descriptions. They also observe the job to become familiar with the necessary skills. Next, trainers meet with subject-matter experts (SMEs), including employees, managers, engineers, or others who are familiar with the job. With the help of these SMEs, trainers identify a list of regularly performed activities and prioritize the list according to importance. Finally, trainers identify the skills and skill levels that are needed to perform the activities or job tasks. For example, nurses must watch for changes in patient conditions, reactions, and comfort levels; they need to identify and recall details when observing patients. These activities require good observation skills, and the trainer needs to find or create a test to measure those skills. Once the skills analysis is complete, trainers conduct a basic (or pretraining) skills evaluation to identify skills gaps that need to be addressed prior to enrolling employees in a training session.

Cognitive Ability

Research shows that cognitive ability influences learning and job performance. **Cognitive ability** includes three dimensions: verbal comprehension, quantitative ability, and reasoning ability.²⁷ *Verbal comprehension* refers to the person's capacity to understand and use written and spoken language. *Quantitative ability* refers to how fast and accurately a person can solve math problems. *Reasoning ability* refers to the person's capacity to invent solutions to problems. Research shows that cognitive ability is related to successful performance in all jobs.²⁸ The importance of cognitive ability for job success increases as the job becomes more complex.

For example, a supermarket cashier needs low to moderate levels of all three dimensions of cognitive ability to perform that job successfully. The supermarket cashier needs to understand the different denominations of bills and coins to give customers the correct amount of change. The cashier also needs to invent solutions

to problems. (For example, how does the cashier deal with items that are not priced that the customer wants to purchase?) Finally, the cashier needs to be able to understand and communicate with customers (verbal comprehension). In comparison, an emergency room physician needs higher levels of verbal comprehension, quantitative ability, and reasoning ability than the cashier. For example, when dealing with an infant experiencing seizures in an emergency situation, the physician needs to be able to calculate the correct dosage of medicine (based on an adult dosage) to stop the seizures after considering the child's weight. The physician also has to be able to diagnose the situation quickly and determine what actions (blood tests, x-rays, respiratory therapy, etc.) are necessary. And finally, the physician needs to communicate clearly with the patient's parents in describing the treatment and recovery process.

Trainees' level of cognitive ability also can influence how well they learn in training programs.²⁹ Trainees with low levels of cognitive ability are more likely to fail to complete training or (at the end of training) receive lower grades on tests that measure how much they have learned.

To identify employees without the cognitive ability to succeed on the job or in training programs, companies use cognitive ability tests. For example, consider the actions taken by the Federal Aviation Administration (FAA) to identify potential air traffic controllers who will complete training successfully.³⁰ Air traffic control work requires quick analytical thinking and strong communications skills. These skills are emphasized and further developed in air traffic controller training. In addition to classroom training, air traffic controllers receive training through computer-based simulations of airport towers and en route centers, which direct planes between airports. The FAA estimates that in the past, it spent \$10 million per year on unsuccessful trainees, which resulted in a doubling of training costs. To reduce its training costs and increase the number of new controllers who will be successful, the FAA uses an eight-hour test of cognitive skills that identifies whether applicants can think spatially, have good short- and long-term memory, and can work well under pressure—skills that are needed by successful air traffic controllers. Determining a job's cognitive ability requirement is part of the task analysis process, discussed later in this chapter.

Reading Ability

Readability refers to the difficulty level of written materials.³¹ An inappropriate reading level can impede performance and learning in training programs. Materials used in training should be evaluated to ensure that their reading level does not exceed that required by the job. A readability assessment usually involves an analysis of sentence length and word difficulty. One estimate is that 1 out of every 10 working-age adults in the United States has limited English-language proficiency.³² An inability to speak and read English prevents workers from advancing to better-paying jobs. Companies who invest in English-language programs can open career paths and help employers retain and promote a diverse workforce.

If trainees' reading level does not match the level required by the training materials, four options are available. First, trainers can determine whether it is feasible to lower the reading level of training materials or use video or on-the-job training, which involves learning by watching and practicing rather than by reading. Second, employees without the necessary reading level could be identified through reading tests and reassigned to other positions more congruent with their skill levels. Third, again using reading tests, trainers can identify employees who lack the necessary reading skills and provide them with remedial training. Fourth, trainers can consider redesigning the job to accommodate employees' reading levels. The fourth option is certainly the most costly and least practical. Therefore, alternative training methods need to be considered, or managers can elect a non-training option. Non-training options include selecting employees for jobs and training opportunities on the basis of reading, computation, writing, and other basic skill requirements.

To develop basic skills or close the skills gap, many companies are engaging in skills assessment, training, or a combination of the two. They are working to identify and close skill gaps, either alone or in partnerships with state government agencies.³³ For example, Agricare, a farm management company, has conducted workplace

English training for three years in Oregon.³⁴ The company planned to expand its classes to its California locations but had to temporarily delay doing so due to the COVID-19 pandemic. The company manages 10,000 acres of farmland and employs 250 people. Its classes are also open to a sister company, Homegrown Organic Farms, which employs 45 people who sell and distribute the fruits that Agricare grows. To help employees improve their English while not being able to attend classes, the company's HR professional is reading through books with some of the employee teams. She is also encouraging employees to download language apps, and the company has purchased and distributed books (in both English and Spanish) to them to help continue building their vocabulary. Managers encourage readiness to learn at Schweitzer Engineering Laboratories, located in the state of Washington.³⁵ The company has a library and offers classes in math, science, and writing. The company hires for problem-solving skills, rather than educational credentials, so workers without a college degree can learn skills as needed to pursue a career path. One such worker is Roy Edwards, who was not motivated by the academic environment he experienced in college but quickly became fascinated by the activities and problems he encountered on the job. He began enrolling in the company-sponsored classes and reading about robotics and now supervises the workers who code them. His story is not uncommon at Schweitzer, where other entry-level employees also have worked their way up to positions as technicians.

PK Controls, an Ohio-based maker of automation controls, has difficulty finding the technicians it needs.³⁶ PK Controls does some of its own training to develop technicians but it also partners with Columbus State Community College to develop employees with the skills it needs. Along with other Ohio companies, including Honda and Worthington Industries, PK Controls participates in Columbus State's Modern Manufacturing Program, which offers an associate degree in electro-mechanical engineering. Students spend two days a week in class and three days a week on the job. On-the-job training starts after the students have completed three semesters of classes, including advanced engineering. Graduates of the program quickly get jobs that pay \$50,000 to \$60,000, and the companies—including PK Controls—get employees with the skills they need.

Self-Efficacy

Self-efficacy is employees' belief that they can perform their job or learn the content of the training program successfully. For many employees who may not have been successful performers in the past, the job environment can be threatening. For example, as you will see in **Chapter Ten**, employees hired through a welfare-to-work program—a program designed to help find jobs for welfare recipients—may lack self-efficacy. The training environment also can be threatening to those workers who have not received training or formal education for some length of time; lack education; or are not experienced in the training program's subject matter. For example, training employees to use equipment for computer-based manufacturing may represent a potential threat, especially if they are intimidated by new technology and lack confidence in their ability to master the skills needed to use a computer. Research has demonstrated that self-efficacy is related to performance in training programs.³⁷ Employees' self-efficacy level can be increased by:

1. Letting employees know that the purpose of training is to try to improve performance rather than to identify areas in which employees are incompetent.
2. Providing as much information as possible about the training program and the purpose of training prior to the actual training.
3. Showing employees the training success of their peers who are now in similar jobs.
4. Providing employees with feedback that learning is under their control and they have the ability and the responsibility to overcome any learning difficulties they experience in the program.

Awareness of Training Needs, Career Interests, and Goals

To be motivated to learn in training programs, employees must be aware of their skill strengths and weaknesses and of the link between the training program and improvement of their weaknesses.³⁸ Managers should make sure that employees understand why they have been asked to attend training programs, and they should communicate the link between training and the improvement of skill weaknesses or knowledge deficiencies. This can be accomplished by sharing performance feedback with employees, holding career development discussions, or having employees complete a self-evaluation of their skill strengths and weaknesses as well as career interests and goals. For example, ESL Federal Credit Union's career counseling services allow employees to proactively manage their careers.³⁹ Using the services, employees can explore departmental career lattices and register for job shadowing, meet with a career coach, and register for micro-internships that provide the opportunity to experience a new role to broaden their skill sets and prepare for a potential future job opportunity.

If possible, employees need to be given a choice of what programs to attend. They should also understand how actual training assignments are made, to maximize their motivation to learn. Several studies have suggested that giving trainees a choice regarding which programs to attend and then honoring those choices maximizes motivation to learn. Giving employees choices but not necessarily honoring them can undermine motivation to learn.⁴⁰

Age

There is biological evidence that certain mental capacities decrease from age 20 to age 70.⁴¹ Short-term memory and the speed at which people process information decline with age. However, with age also comes experience, which can compensate for the loss of memory and mental quickness. Although mental quickness and memory losses diminish at a steady pace, at older ages, memory loss is much greater because mental resources are more depleted than at earlier ages. As we discuss in **Chapter Five**, there is some evidence that certain type of training methods are more beneficial for older trainees.⁴²

Input

Employees' perceptions of two characteristics of the work environment—situational constraints and social support—are determinants of performance and motivation to learn. **Situational constraints** include lack of proper tools and equipment, materials and supplies, budgetary support, and time. **Social support** refers to managers' and peers' willingness to provide feedback and reinforcement.⁴³ If employees have the knowledge, skills, attitudes, and behavior needed to perform but do not have the proper tools and equipment needed, their performance will be inadequate.

To ensure that the work environment enhances trainees' motivation to learn, managers should take the following steps:

1. Provide materials, time, job-related information, and other work aids necessary for employees to use new skills or behavior before participating in training programs.
2. Speak positively about the company's training programs to employees.
3. Let employees know they are doing a good job when they are using training content in their work.
4. Encourage work-group members to involve each other in trying to use new skills on the job by soliciting feedback and sharing training experiences and situations in which training content has been helpful.
5. Provide employees with time and opportunities to practice and apply new skills or behaviors to their work.

Output

Poor or substandard performance can occur on the job because employees do not know at what level they are expected to perform. For example, they may not be aware of quality standards related to speed or the degree of personalization of service that is expected. Employees may have the knowledge, skills, and attitudes necessary to perform and yet fail to perform because they are not aware of the performance standards. Lack of awareness of the performance standards is a communications problem, but it is not a problem that training can “fix.”

Understanding the need to perform is important for learning. Trainees need to understand what specifically they are expected to learn in the training program. To ensure that trainees master training content at the appropriate level, trainees in training programs also need to understand the level of proficiency that is expected of them. For example, for tasks, level of proficiency relates to how well employees are to perform a task. For knowledge, level of proficiency may relate to a score on a written test. The standards or the level of performance is part of the learning objectives (discussed in **Chapter Four**).

Consequences

If employees do not believe that rewards or incentives for performance are adequate, they will be unlikely to meet performance standards even if they have the necessary knowledge, behaviors, skills, or attitudes. Also, work-group norms may encourage employees to not meet performance standards. **Norms** refer to accepted standards of behavior for work-group members. For example, during labor contract negotiations, airline baggage handlers for a major airline worked slowly loading and unloading baggage from airplanes. As a result, many passenger departures and arrivals were delayed. The baggage handlers had the knowledge, skills, and behaviors necessary to load and unload the planes more quickly, but they worked slowly because they were trying to send a message to management that the airlines could not perform effectively if their contract demands were not met.

Consequences also affect learning in training programs. Incentive systems, such as providing gift cards redeemable for food, clothes, or movies or accumulating points that can be used toward paying for enrollment in future courses, may be useful for motivating some employees to attend and complete training courses (discussed in **Chapter Five**).⁴⁴ However, one of the most powerful ways to motivate employees to attend and learn from training is to communicate the personal value of the training. For example, how will it help them improve their skills or career? How will it help them deal with problems they encounter on the job? It is important that the communication from the manager about potential benefits be realistic. Unmet expectations about training programs can hinder the motivation to learn.⁴⁵

Feedback

Performance problems can result when employees do not receive feedback regarding the extent to which they are meeting performance standards. Employees may know what they are supposed to do (output), but they may not understand how close their performance is to the standard. Training may not be the best solution to this type of problem. Employees need to be given specific, detailed feedback regarding both effective and ineffective performance. For employees to perform to standard, feedback needs to be given frequently, not just during a yearly performance evaluation.

In **Chapter Four**, the role of feedback in learning is discussed in detail. Keep in mind that feedback is critical for shaping trainees’ behaviors and skills.

Determining Whether Training Is the Best Solution

A root cause analysis is often used to determine whether training is the best solution. A **root cause analysis** refers to the process of determining whether training is the best or most likely solution to a performance problem or gap. There are four methods that are useful for conducting a root cause analysis.⁴⁶ First, the following seven questions should be considered:

1. Is the performance problem important? Does it have the potential to cost the company a significant amount of money from lost productivity or customers?
2. Do employees know how to perform effectively? Perhaps they received little or no previous training, or the training was ineffective. (This problem is a characteristic of the person.)
3. Can employees demonstrate the correct knowledge or behavior? Perhaps employees were trained but they infrequently or never used the training content (knowledge, skills, etc.) on the job. (This is an input problem.)
4. Were performance expectations clear (input)? Were there any obstacles to performance, such as faulty tools or equipment?
5. Were positive consequences offered for good performance? Was good performance not rewarded? For example, when employees are dissatisfied with their compensation, their peers or their union may encourage them to slow down their pace of work. (This involves consequences.)
6. Did employees receive timely, relevant, accurate, constructive, and specific feedback about their performance (a feedback issue)?
7. Were other solutions—such as job redesign or transferring employees to other jobs—too expensive or unrealistic?

If employees lack the knowledge and skills to perform a job and the other factors are satisfactory, training is needed. If employees have the knowledge and skills to perform but input, output, consequences, and/or feedback are inadequate, training may not be the best solution. For example, if poor performance results from faulty equipment, training cannot solve this problem, but repairing the equipment will. If poor performance results from lack of feedback, then employees may not need training, but their managers may need training on how to give performance feedback.

Second, the 5 Whys, another questioning method, can be used to explore the root cause of a problem. Examples of possible questions include, “Why are employees performing poorly?” or “Why can’t employees greet customers as they enter the store?” The questions are used to identify possible causes that can be acted upon. For each cause identified, you need to consider if it is occurring because of a lack of knowledge, skills, or behaviors that can be influenced by training or if other factors are responsible. Third, the Fishbone diagram can be used. The Fishbone diagram is a structured brainstorming technique that can be used to help identify, explore, and visually display the possible causes of a problem. It considers how machines, methods, materials, measurements, the environment, and people contribute to a problem. Fourth, the 4 Square method involves drawing a square with four different themes represented: motivation, resources, training, and bad job fit. The factors that are believed to be causing the problem can be listed in the appropriate theme. The 4 Square method is useful in identifying whether training alone is the problem or if motivation, resources, and/or bad job fit are also involved. For example, if an employee knows how to perform well but there is a lack of motivation to do so, then lack of training is not the likely cause of the problem. However, if an employee doesn’t know how to perform well then the lack of training is likely a probable cause of the problem.

It is also important to consider the relationships among a critical job issue (a problem or opportunity that is critical to the success of a job within the company), a critical process issue (a problem or opportunity that is critical to the success of a business process), and a critical business issue (a problem or opportunity that is

critical to the success of the company).⁴⁷ If the critical job issue, critical process issue, and critical business issue are related, training should be a top priority because it will have a greater effect on business outcomes and results and will likely receive greater management support. **Table 3.5** shows the relationships among the critical job, process, and business issues for a sales representative. This analysis resulted from a request from a top manager who suggested that sales representatives needed more training because incomplete sales orders were being submitted to production.

TABLE 3.5 Example of the Relationships Among a Critical Job Issue, a Critical Process Issue, and a Critical Business Issue

Critical Job Issue	Critical Process Issue	Critical Business Issue
<i>Desired Results</i>	<i>Desired Results</i>	<i>Desired Results</i>
No incomplete order forms 100 percent accurate orders	Order cycle time of 3 days	Market share of 60 percent
<i>Current Results</i>	<i>Current Results</i>	<i>Current Results</i>
10 percent incomplete order forms 83 percent accurate orders	Order cycle time of 30 days	Market share of 48 percent

Source: Based on G. A. Rummler and K. Morrill, "The Results Chain," *T+D*, February 2005, pp. 27–35.

Papa Murphy's International, the franchise pizza store, conducted a root cause analysis to identify gaps in employees' knowledge, skills, and attitudes.⁴⁸ The analysis identified the need to provide employees, managers, and franchise owners with small chunks of learning available on an as-needed basis. Based on this analysis, the company developed a four tier learning approach. Tier 1 includes the knowledge and skills every new employees needs when they start working in a store. Tier 2 focuses on leadership, time management, and organizational skills needed by shift supervisors and assistant managers. Tier 3 focuses on the skills and knowledge needed to be an effective store manager. Tier 4 emphasizes the skills franchise owners and operators need to be successful. Consider the process a global retailer of rugged, athletic, casual, and stylish clothes for kids and young adults uses to identify whether a performance need should be addressed by training or other solutions. **Figure 3.4** shows the process this retailer uses to determine the type of need, who is affected, and the corrective strategy. As you can see from **Figure 3.4**, training and development is only one of several possible solutions and addresses issues related specifically to knowledge gaps, sharing knowledge and skills, informal learning, and manager and supervisor support.

Task Analysis

Task analysis results in a description of work activities, including tasks performed by the employee and the knowledge, skills, and abilities required to complete the tasks. A **job** is a specific position requiring the completion of certain tasks. A **task** is the employee's work activity in a specific job. **Table 3.6** shows several tasks for the territory manager job. To complete these tasks, employees must have specific levels of knowledge, skill, ability, and other considerations (KSAOs). **Knowledge** includes facts or procedures. **Table 3.8** later in this section shows examples of the knowledge needed for tasks for the territory manager job. **Skill** indicates competency in performing a task. For the territory manager job, analyzing data and information is necessary for identifying underlying principles, reasons, or facts, making appropriate recommendations or decisions, and anticipating problems, opportunities, and needs of customers. **Ability** includes the physical and mental capacities to perform a task. One of the abilities for the territory manager's job is stress tolerance, which involves accepting criticism and dealing calmly and effectively with high stress situations. **Other** refers to the

FIGURE 3.4 Identifying How to Solve Performance Issues

1. WHAT?

DETERMINE THE TYPE OF NEED:

PROBLEM

IMPROVEMENT

FUTURE PLANNING

2. WHO?

DETERMINE THE LEVEL OF THE ORGANIZATION IMPACTED:

ENTIRE
ORGANIZATION

DIVISION

DEPARTMENT

INDIVIDUAL

JOB

3. HOW?

DETERMINE THE CORRECTIVE STRATEGY:

IS THERE A SYSTEMS PROBLEM?

ARE THERE PROPER SYSTEMS AND RESOURCES
AVAILABLE FOR PERFORMANCE TO IMPROVE?

ARE WE RECRUITING THE RIGHT PEOPLE?

ARE OUR RECRUITING EFFORTS TARGETED TO ATTRACT
EMPLOYEES WHO START OUT WITH THE
BASIC COMPETENCIES TO DO THE JOB?

IS COACHING NEEDED?

DOES INFORMAL TRAINING OR PASSING OF INFORMATION
EXIST? IS IT SUPPORTED? ARE THERE RESOURCES FOR
ASSOCIATES BUILT INTO THEIR ROLES?IS THERE AN ORGANIZATIONAL
DEVELOPMENT PROBLEM?ARE THERE POLICY ROADBLOCKS TO PERFORMANCE?
DO INCENTIVES MATCH PERFORMANCE?

IS PLACEMENT AN ISSUE?

IS THE WRONG ASSOCIATE IN THE JOB? HAVE YOU
EXPLORED THE BEHAVIORAL TENDENCIES OF ASSOCIATES?
DO THEY MATCH THEIR JOB REQUIREMENTS?

IS THERE A TRAINING NEED?

DO ASSOCIATES HAVE THE POTENTIAL TO PERFORM JOB
FUNCTIONS? IS THERE A KNOWLEDGE GAP THAT NEEDS
TO BE FILLED? ARE MANAGERS AND SUPERVISORS
CREATING A SUPPORTIVE ENVIRONMENT THAT
ENCOURAGES PERFORMANCE?

conditions under which tasks are performed. These conditions include identifying the equipment involved and the environment in which the employee works (e.g., the need to wear an oxygen mask and work in extremely hot conditions), time constraints for a task (e.g., deadlines), safety considerations, or performance standards. For the territory manager, an important condition is willingness to travel to different locations within a state or territory.

Task analysis should be undertaken only after the organizational analysis has determined that the company wants to devote time and money for training. Why? Task analysis is a time-consuming, tedious process that involves a large time commitment to gather and summarize data from many different persons in the company, including managers, job incumbents, and trainers.

Steps in a Task Analysis

A task analysis involves four steps:⁴⁹

1. Select the job or jobs to be analyzed.
2. Develop a preliminary list of tasks performed on the job by (1) interviewing and observing expert employees and their managers and (2) talking with others who have performed a task analysis.
3. Validate or confirm the preliminary list of tasks. This step involves having a group of SMEs (job incumbents, managers, etc.) answer several questions regarding the tasks, either in a meeting or on a written survey. The types of questions that may be asked include the following: How frequently is the task performed? How much time is spent performing each task? How important or critical is the task for successful performance of the job? How difficult is the task to learn? Is performance of the task expected of entry-level employees?

Table 3.6 presents a sample of part of a task analysis questionnaire. This information is used to determine which tasks will be focused on in the training program. The person or committee conducting the needs assessment must decide the level of ratings across dimensions that will determine that a task should be included in the training program. Tasks that are important, frequently performed, and of moderate-to-high level of difficulty are tasks for which training should be provided. Tasks that are not important or are infrequently performed should not involve training. Based on understanding the job and the implications of errors or mistakes in completing tasks, managers and trainers need to decide if tasks that are important but performed infrequently and require minimal difficulty should be included in training. Also, managers and trainers must determine whether or not important tasks—regardless of how frequently they are performed or their level of difficulty—will be included in training.

TABLE 3.6 Sample Task Analysis Questionnaire

Name	Date
Position	
Please rate each of the task statements according to three factors: (1) the <i>importance</i> of the task for effective performance, (2) how <i>frequently</i> the task is performed, and (3) the degree of <i>difficulty</i> required to become effective in the task. Use the following scales in making your ratings.	
<i>Importance</i>	<i>Frequency</i>
4 = Task is critical for effective performance.	4 = Task is performed once a day.
3 = Task is important but not critical for effective performance.	3 = Task is performed once a week.
2 = Task is of some importance for effective performance.	2 = Task is performed once every few months.
1 = Task is of no importance for effective performance.	1 = Task is performed once or twice a year.
0 = Task is not performed.	0 = Task is not performed.
<i>Difficulty</i>	
4 = Effective performance of the task requires extensive prior experience and/or training (12–18 months or longer).	

continued

3 = Effective performance of the task requires minimal prior experience and training (6–12 months).			
2 = Effective performance of the task requires a brief period of prior training and experience (1–6 months).			
1 = Effective performance of the task does not require specific prior training and/or experience.			
0 = This task is not performed.			
<i>Task</i>	<i>Importance</i>	<i>Frequency</i>	<i>Difficulty</i>
1. Manage the production and profitability with all agencies in the assigned sales territory 2. Understand the agents' business, including their sales, support, and service strategies 3. Recommend company products to the agents expertly 4. Establish, implement, and monitor annual business plans for agencies in the assigned sales territory 5. Develop and maintain relationships with agencies; manage their expectation and help them when needed			

4. Once the tasks have been identified, it is important to identify the knowledge, skills, or abilities that are difficult to learn or prone to errors, such as those required for decision-making or problem-solving tasks. For these tasks it is necessary to determine how the thought processes of experts differ from those of novices. This information is useful for designing training that includes the right amount of practice and feedback for novices to learn. This information can be collected through interviews and questionnaires. Recall this chapter's discussion of how ability influences learning. Information concerning basic skill and cognitive ability requirements is critical for determining if certain levels of knowledge, skills, and abilities will be prerequisites for entrance to the training program (or job) or if supplementary training in underlying skills is needed. For training purposes, information concerning how difficult it is to learn the knowledge, skill, or ability is important—as is whether the knowledge, skill, or ability is expected to be acquired by the employee before taking the job.⁵⁰

Table 3.7 summarizes key points to remember regarding task analysis.

TABLE 3.7 Key Points to Remember When Conducting a Task Analysis

- A task analysis should identify both what employees are actually doing and what they should be doing on the job.
- Task analysis begins by breaking the job into duties and tasks.
- Use more than two methods for collecting task information to increase the validity of the analysis.

- For task analysis to be useful, information needs to be collected from SMEs, including job incumbents, managers, and employees familiar with the job.
- In deciding how to evaluate tasks, the focus should be on tasks necessary to accomplish the company's goals and objectives. These may not be the tasks that are the most difficult or take the most time.

Sources: Based on A. P. Carnevale, L. J. Gainer, and A. S. Meltzer, *Workplace Basics Training Manual* (San Francisco: Jossey-Bass, 1990); E. A. Surface, "Training Need Assessment: Aligning Learning and Capability with Performance Requirements and Organizational Objectives," in *The Handbook of Work Analysis: Methods, Systems, Applications and Science of Work Measurement in Organizations*, 1st ed., eds. M. A. Wilson, W. Bennett, S. G. Gibson, and M. Alliger (Routledge Academic, 2012), pp. 437–62.

Example of a Task Analysis

Each of the four steps of a task analysis can be seen in the following example from an insurance company in **Table 3.8**.⁵¹ First, the territory manager job was the focus of the task analysis. The reason this job was chosen was because of the need to identify tasks and KSAOs that would serve as the basis for the training program objectives and lesson plans.

TABLE 3.8 Sample of a Task-KSAO Grid for Territory Manager

		Manage the production and profitability with all agencies in the assigned sales territory	Understand the agents' business, including their sales, support, and service strategies	Recommend Grange products to the agents expertly	Establish, implement, and monitor annual business plans for agencies in the assigned sales territory
KNOWLEDGE		T1	T2	T3	T4
Knowledge of company products	K1	X		X	
Knowledge of insurance industry / rules and policies	K2		X	X	
Knowledge of Customer Relation Management (CRM) software	K3	X	X		X
Knowledge of IBM On-Demand program	K4	X	X		X

The second step involved interviews with SMEs (managers, job incumbents). The purpose of the interviews was to identify the job's main duties and responsibilities, tools, software, or other aid needed to complete each duty, how success in the job was determined, records or reports prepared, and level of responsibility, autonomy, and accountability. Based on this information, a list of tasks was developed. SMEs reviewed the

list to categorize the tasks by area of job responsibility. The tasks were grouped into five areas of responsibility: agency management, coordination with profit center, market analysis, external relations, and administration. For example, the task “Manage the production and profitability with all agencies in the assigned sales territory” was considered a responsibility of agency management. The job responsibilities and task list were reviewed by the SMEs to ensure that it was complete and accurate.

The third step involved developing a questionnaire including the tasks. The questionnaire was administered to all territory managers and their bosses. The questionnaire included 19 tasks. **Table 3.6** shows a sample from the questionnaire. Territory managers and their bosses were asked to rate each task on importance, frequency, and difficulty. The primary requirement used to determine whether a task required training was its importance rating. A task rated “very important” was identified as one requiring training regardless of its frequency or difficulty. If a task was rated moderately important but difficult, it also was designated for training. Tasks rated as unimportant, not difficult, or done infrequently were not designated for training.

The fourth step included identifying the knowledge, skills, abilities, and other important characteristics needed to complete the tasks. Focus groups with the SMEs identified nine different types of knowledge (e.g., knowledge of the company’s insurance products, computer software, sales and marketing) and twelve skills (e.g., analyzing data and other information, verbal and written communications) abilities (e.g., stress tolerance), and other characteristics (e.g., willingness to travel). A grid showing each task and the knowledge, skills, abilities, and other characteristics required to complete it was developed. **Table 3.8** shows a sample of part of the grid. An “X” indicates the KSAO shown is necessary for the task.

The tasks that were identified based on the ratings as designated for training, and their corresponding area of job responsibility were used to develop course objectives and lesson plans. The tasks were also reviewed to identify prerequisite or required KSAOs for the tasks and area of job responsibility.

Competency Models

In today’s global and competitive business environment, many companies are finding that it is difficult to determine whether employees have the capabilities needed for success. The necessary capabilities may vary from one business unit to another and even across roles within a business unit. As a result, many companies are using competency models to help them identify the knowledge, skills, and personal characteristics (attitudes, personality) needed for successful performance in a job. Competency models are also useful for ensuring that training and development systems are contributing to the development of such knowledge, skills, and personal characteristics.

Traditionally, needs assessment has involved identifying knowledge, skills, abilities, and tasks. However, a current trend in training is for needs assessment to focus on competencies, especially for managerial positions. **Competencies** are sets of skills, knowledge, abilities, and personal characteristics that enable employees to perform their jobs successfully.⁵²

A **competency model** identifies the competencies necessary for each job. Competency models provide descriptions of competencies that are common for an entire occupation, organization, job family, or specific job. Competency models can be used for performance management. However, one of the strengths of competency models is that they are useful for a variety of human resource (HR) practices, including recruiting, selection, training, and development. Competency models can be used to help identify the best employees to fill open positions and to serve as the foundation for development plans that allow employees and their manager to target specific strengths and development areas. The competencies included in competency models vary according to the company’s business strategy and goals. They can include sales, leadership, interpersonal, technical, and other types of competencies. Competency models typically include the name of each

competency, the behaviors that represent proficiency in the competency, and levels that include descriptions representing demonstrated levels of mastery or proficiency. **Table 3.9** shows the technical cluster of competencies from a competency model for a systems engineer. The left side of the table lists technical competencies within the technical cluster (such as systems architecture, data migration, and documentation). The right side shows behaviors that might be used to determine a systems engineer's level of proficiency for each competency.

TABLE 3.9 Competencies from a Competency Model

Technical Cluster	Proficiency Ratings
Systems Architecture Ability to design complex software applications, establish protocols, and create prototypes.	0 —Is not able to perform basic tasks. 1 —Understands basic principles; can perform tasks with assistance or direction. 2 —Performs routine tasks with reliable results; works with minimal supervision. 3 —Performs complex and multiple tasks; can coach or teach others. 4 —Considered an expert in this task; can describe, teach, and lead others.
Data Migration Ability to establish the necessary platform requirements to efficiently and completely coordinate data transfer.	0 —Is not able to perform basic tasks. 1 —Understands basic principles; can perform tasks with assistance or direction. 2 —Performs routine tasks with reliable results; works with minimal supervision. 3 —Performs complex and multiple tasks; can coach or teach others. 4 —Considered an expert in this task; can describe, teach, and lead others.
Documentation Ability to prepare comprehensive and complete documentation, including specifications, flow diagrams, process control, and budgets.	0 —Is not able to perform basic tasks. 1 —Understands basic principles; can perform tasks with assistance or direction. 2 —Performs routine tasks with reliable results; works with minimal supervision. 3 —Performs complex and multiple tasks; can coach or teach others. 4 —Considered an expert in this task; can describe, teach, and lead others.

Source: R. J. Mirabile, "Everything You Wanted to Know About Competency Modeling," *Training and Development*, August 1997, pp. 73-77.

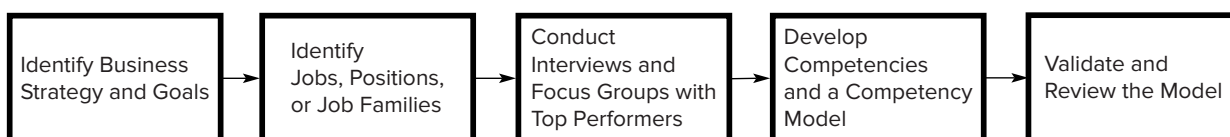
One way to understand competency models is to compare them to job analysis. As you may recall from other classes or experiences, **job analysis** refers to the process of developing a description of the job (tasks, duties, and responsibilities) and the specifications (knowledge, skills, and abilities) that an employee must have to perform it. How does job analysis compare to competency models? Job analysis is more work- and task-focused (what is accomplished), whereas competency modeling is worker-focused (how objectives are met or how work is accomplished). Focusing on “how” versus “what” provides valuable information for training and development. A recent study asked competency modeling experts (consultants, HR practitioners, academics, and industrial psychologists) to compare and contrast competency modeling and job analysis.⁵³ The study found several differences between job analysis and competency models. Competency models are more likely to link competencies and the company’s business goals. Competency models provide descriptions of competencies that are common for an entire occupational group, level of jobs, or an entire organization. In contrast, job analysis describes what is different across jobs, occupational groups, or organization levels and generates specific knowledge, skills, and abilities for particular jobs. It is used to generate specific requirements to be used for employee selection. The competencies generated by competency modeling are more general and believed to have greater application to a wider variety of purposes, including selection, training, employee development, and performance management.

Another way to think about competency models is by considering performance management.⁵⁴ Unfortunately, many performance management systems suffer from a lack of agreement on what outcomes should be used to evaluate performance. Manager–employee discussions about performance deficiencies tend to lack specificity. By identifying the areas of personal capability that enable employees to perform their jobs successfully, competency models ensure an evaluation of what gets done and how it gets done. Performance feedback can be directed toward specific concrete examples of behavior, and knowledge, skills, ability, and other characteristics that are necessary for success are clearly described.

How are competencies identified and competency models developed? **Figure 3.5** shows the process used to develop a competency model. First, the business strategy is identified. The implications of business strategy for training were discussed in **Chapter Two**. The business strategy helps identify what types of competencies are needed to ensure that business goals are met and the company’s strategy is supported. Changes in the business strategy might cause new competencies to be needed or old competencies to be altered. Second, the job or position to be analyzed is identified. Third, effective and ineffective performers are identified. Fourth, the competencies responsible for effective and ineffective performance are identified. There are several approaches for identifying competencies. These include analyzing one or several “star” performers, surveying persons who are familiar with the job (SMEs), and investigating benchmark data of good performers in other companies.⁵⁵ Fifth, the model is validated. That is, a determination is made as to whether the competencies included in the model truly are related to effective performance. In the example of the technical competencies for the systems engineer shown in **Table 3.9**, it is important to verify that (1) these three competencies are necessary for job success, and (2) the level of proficiency of the competency is appropriate.

Following the development process outlined in **Figure 3.5** will ensure that competencies and competency models are valid. However, trainers, employees, managers, and other experts should be trained (especially inexperienced raters) in how to determine accurate competency ratings. Training should ensure that raters

FIGURE 3.5 The Process Used in Developing a Competency Model



understand each competency and the differences between them and can distinguish between low, medium, and high levels of proficiency.⁵⁶

Competency models are useful for training and development in several ways:⁵⁷

- They identify behaviors needed for effective job performance. These models ensure that feedback given to employees as part of a development program (such as 360-degree feedback) relate specifically to individual and organizational success.
- They provide a tool for determining what skills are necessary to meet today's needs, as well as the company's future skill needs. They can be used to evaluate the relationship between the company's current training programs and present needs. That is, they help align training and development activities with the company's business goals. They can be used to evaluate how well the offerings relate to anticipated future skill needs.
- They help determine what skills are needed at different career points.
- They provide a framework for ongoing coaching and feedback to develop employees for current and future roles. By comparing their current personal competencies to those required for a job, employees can identify competencies that need development and choose actions to develop those competencies. These actions may include courses, job experiences, and other types of development. (Development methods are detailed in **Chapter Nine**, "Employee Development and Career Management.")
- They create a "road map" for identifying and developing employees who may be candidates for managerial positions (succession planning).
- They provide a common set of criteria that may be used for identifying appropriate development training and learning activities for employees, as well as for evaluating and rewarding them. This helps integrate and align the company's HR systems and practices.

For example, the talent development team at Providence St. Joseph Health recruits and develops nursing professionals in a highly competitive environment due to a nationwide nursing shortage and a high first year turnover rate.⁵⁸ To attract new talent and improve retention, the hospital system created the Nursing Institute to assess workforce and industry trends and develop strategies to address employee needs. The Nursing Institute is working to ensure all new employees meet important benchmarks in a systematic manner. To do so, the talent development team has developed a standardized competency assessment form for new graduate registered nurses. Previously each hospital across the Providence St. Joseph system had been managing competencies individually, with great variation across the system and among specialties. The new standardized process assesses nurses in 14 areas and is intended for use by a registered nurse's supervisor. The tool can also be used to define goals and demonstrate level of development of each competency during a nurse's orientation period.

Table 3.10 shows the competency model used by Voya Financial, a company that specializes in retirement planning and investment.⁵⁹ These competencies are used in performance management to assess how employees achieved their results and serve as the basis for the company's leadership development programs. To use competency models effectively for performance evaluation, they must be up to date, drive business performance, be job-related (valid), be relevant (or customized) for all of the company's business units, and provide sufficient detail to make an accurate assessment of employees' performance.

TABLE 3.10 Examples of Competencies at Voya Financial

Demonstrate Integrity
Focus on the Customer
Build a Strong Organization

continued

Create & Lead Change
Deliver Results
Excel at Execution
Lead with Passion & Clarity
Develop People & Self
Work as One Company

Source: From “Voya’s Leadership Model,” Voya, from www.voya.com/page/why-voya, accessed February 2, 2021.

Scope of Needs Assessment

Up to this point, the chapter has discussed the various aspects of needs assessment, including organizational, person, and task analyses. This involves interviews, observations, and potentially even surveying employees. You might be saying to yourself, “This sounds good, but it appears to be a very elaborate process that takes time. What happens if I don’t have time to conduct a thorough needs assessment? Should I abandon the process?”

Needs assessment is often skipped for several reasons based on assumptions such as training is always the issue or is mandated; it’s too costly, takes too long, and is too complex; and managers will not cooperate. Despite the constraints to conducting a needs assessment, it is necessary to determine if a problem or pressure point exists and to identify the best solution, which could be training.

However, even if managers demand a training course right now, needs assessment should still be conducted. There are several ways to conduct a rapid needs assessment. A **rapid needs assessment** refers to a needs assessment that is done quickly and accurately, but without sacrificing the quality of the process or the outcomes.⁶⁰ The key to conducting a rapid needs assessment is choosing the needs assessment methods that will provide the results you can have the greatest confidences in while using the fewest resources (i.e., time, money, SMEs). There are several ways to conduct a rapid needs assessment. First, the scope of needs assessment depends on the size of the potential pressure point. If the pressure point seems to be local and has a potentially small impact on the business, then the information-gathering part of needs assessment could consist of only a few interviews with managers or job incumbents. If the pressure point will have a large impact on the business, then more information gathering should be conducted. If, after interviewing SMEs and job incumbents, you can tell that you are not learning anything new about the job, then interviewing could be stopped. Second, consider using already available data collected for other purposes. Error data, sales data, customer complaints, and exit interviews might provide valuable clues as to the source of performance and survey problems. JetBlue uses on-the-job performance data and business results data to identify training needs.⁶¹ For example, using data collected when aircraft are regularly serviced, JetBlue found that there was an increase in cosmetic damage to airplanes. This triggered the learning team to conduct more in-depth assessment to identify potential learning needs that may have resulted in an increase in the damage rate over time. Customer complaints tracked by the U.S. Transportation Department revealed an increase in problems experienced by JetBlue’s disabled passengers before boarding. Based on these data, training was revised and expanded, resulting in fewer complaints. The web may be another useful source for quickly conducting interviews and surveys with SMEs in different locations. Finally, if you are attuned to the business problems, technological developments, and other issues facing the organization, you will be able to anticipate training needs. For example, if the company is opening sales offices in an international location and introducing new technology in the manufacturing plants, cross-cultural training and training designed to help employees use the new technology undoubtedly will be needed. Be prepared by understanding the business.

Needs Assessment in Practice

KLA-Tencor supplies process controls and equipment to the semiconductor industry.⁶² KLA-Tencor service engineers diagnose and repair complex machines that use advanced laser, optical, and robotic technologies. The engineers need to maintain proficiency in their current skills as well as add new skills to keep pace with new technology used in the company's equipment. This is critical for KLA-Tencor to quickly solve equipment problems, which, if unresolved, can result in millions of dollars of lost revenue for its customers. Providing effective service is critical for the company to keep current customers and develop new business. In fact, one of the company's values is "Indispensable" (the other values are "Perseverance," "Drive to Be Better," "High Performance Teams," and "Honest, Forthright and Consistent").

KLA-Tencor uses a skill management process (the Right People, Right Knowledge process) to monitor its workforce skills and uses this information to change its training programs. The process involves developing a task list, training on the task, practicing on-the-job training to gain certification, and conducting an annual skills assessment. To conduct the skills assessment, a survey was sent to all of KLA-Tencor's more than 1,000 service engineers. For each task, the engineers were asked to rate their capability of doing the task on a scale from "I don't know how" to "I can teach it to others." Also, they were asked to evaluate how frequently they performed the task, from "Never" to "More than two times per year." Based on their responses, they were assigned a training task. More than 200 courses were created to train the engineers. To ensure that the training was completed, both engineers and their managers were held accountable. This helped achieve a 95 percent completion rate within one year after training was assigned. The skills assessment data were also used to identify gaps in current training, resulting in more than 2,000 changes in courses and certification programs. The skills assessment is done annually to ensure that service engineers keep up to date with new technology and products.

This example illustrates several aspects of the needs assessment process. First, training was viewed as critical for helping the company meet its strategic objectives. As a result, resources and time were allocated for needs assessment and training. Second, the needs assessment included a task or skill assessment that helped determine who needed training and what tasks they needed to learn. Third, based on the needs assessment, training programs were developed or changed to improve the identified skill deficiencies.

Summary

The first step in a successful training effort is to determine that a training need exists through a process known as needs assessment. Needs assessment involves three steps: organizational analysis, person analysis, and task analysis. Various methods—including observation, interviews, and surveys or questionnaires—are used to conduct a needs assessment. Each has advantages and disadvantages. Organizational analysis involves determining (1) the extent to which training is congruent with the company's business strategy and resources and (2) if peers and managers are likely to provide the support needed for trainees to use training content in the work setting.

Person analysis focuses on identifying whether there is evidence that training is the solution, who needs training, and whether employees have the prerequisite skills, attitudes, and beliefs needed to ensure that they master the content of training programs. Because performance problems are one of the major reasons that companies consider training for employees, it is important to investigate how personal characteristics, input, output, consequences, and feedback relate to performance and learning. Managers and trainers need to be concerned about employees' basic skill levels, attitudes, age and generation, and the work environment in determining if performance problems can be solved using training and how training should be designed.

144 PART TWO: Designing Training

Training is likely the best solution to a performance problem if employees don't know how to perform. If employees have not received feedback about their performance, if they lack the equipment needed to perform the job, if the consequences for good performance are negative, or if they are unaware of an expected standard for performance, then training is not likely to be the best solution.

To maximize employees' motivation to learn in training programs, managers and trainers need to understand these factors prior to sending employees to training. For example, lack of basic skills or reading skills can inhibit both job performance and learning.

A task analysis involves identifying the task and the training that employees will require in terms of knowledge, skills, and abilities. Competency modeling is a new approach to needs assessment that focuses on identifying personal capabilities, including knowledge, skills, attitudes, values, and personal characteristics.

Key Terms

needs assessment 113	person characteristics 124	root cause analysis 132
organizational analysis 113	input 124	job 133
person analysis 113	output 124	task 133
task analysis 113	consequences 124	knowledge 133
stakeholders 115	feedback 124	skill 133
learning council 116	motivation to learn 126	ability 133
subject-matter expert (SME) 117	basic skills 126	other 133
job incumbent 117	cognitive ability 127	competencies 138
focus group 119	readability 128	competency model 138
crowdsourcing 119	self-efficacy 129	job analysis 140
benchmarking 121	situational constraints 130	rapid needs assessment 142
gap analysis 124	social support 130	
readiness for training 124	norms 131	

Discussion Questions

1. Which of the factors that influence performance and learning do you think is most important? Which is least important?
2. If you had to conduct a needs assessment for a new job at a new plant, describe the method you would use.
3. If you were going to use technology to identify training needs for customer service representatives for an e-commerce clothing company, what steps would you take to ensure that the technology was not threatening to employees?
4. Needs assessment involves organization, person, and task analyses. Which one of these analyses do you believe is most important? Which is least important? Why?

5. Why should upper-level managers be included in the needs assessment process?
6. Explain how you would determine if employees had the reading level necessary to succeed in a training program. How would you determine if employees had the necessary computer skills needed to use an online training program?
7. What conditions would suggest that a company should buy a training program from an outside vendor? Which would suggest that the firm should develop the program itself?
8. Assume that you have to prepare older employees with little computer experience to attend a training course on how to use the Internet. How will you ensure that they have high levels of readiness for training? How will you determine their readiness for training?
9. Explain the process you would use to conduct a root cause analysis of a performance problem.
10. Review the sample tasks and task ratings for the electronic technician's job shown next. What tasks do you believe should be emphasized in the training program? Why?

Task	Importance	Frequency	Learning Difficulty
1. Replaces components	1	2	1
2. Repairs equipment	2	5	5
3. Interprets instrument readings	1	4	5
4. Uses small tools	2	5	1

Explanation of ratings:

Frequency: 1 = very infrequently to 5 = very frequently

Importance: 1 = very important to 5 = very unimportant

Learning difficulty: 1 = easy to 5 = very difficult

11. Why would we consider age and generational differences as part of a needs assessment? Is this important? Explain.
12. Discuss the types of evidence that you would look for to determine whether a needs analysis has been conducted improperly.
13. How is competency modeling similar to traditional needs assessment? How does it differ?
14. What is a rapid needs assessment? How would you conduct a rapid needs assessment so that it is valuable and accurately identifies training needs?



Application Assignments

1. Develop a competency model for a job held by a friend, spouse, or roommate (someone other than yourself). Use the process discussed in this chapter to develop your model. Note the most difficult part of developing the model. How could the model be used?
2. The Department of Social Services represents a large portion of your county's budget and total number of employees. The job of eligibility technician is responsible for all client contact, policy interpretation, and financial decisions related to several forms of public aid (e.g., food stamps, aid to families with dependent children). Eligibility technicians must read a large number of memos and announcements of new and revised policies and procedures. Eligibility technicians were complaining that they had

difficulty reading and responding to this correspondence. The county decided to send the employees to a speed reading program costing \$500 per person. The county has 200 eligibility technicians.

Preliminary evaluation of the speed reading program was that trainees liked it. Two months after the training was conducted, the technicians told their managers that they were not using the speed reading course in their jobs, but were using it in leisure reading at home. When their managers asked why they weren't using it on the job, the typical response was, "I never read those memos and policy announcements anyway."

- a. Evaluate the needs assessment process used to determine that speed reading was necessary. What was good about it? Where was it faulty?
 - b. How would you have conducted the needs assessment? Be realistic.
3. Consider the interview questions for the lead drivers that are shown on page 133. Write questions that could be used to interview the six lead driver supervisors and the two regional vice presidents. How do these questions differ from those for the lead drivers? How are they similar?
4. Several companies are known for linking their mission, values, and HR practices in ways that have led to business success as well as employee satisfaction. These companies include Southwest Airlines (www.iflyswa.com), Cisco Systems (www.cisco.com), SAS Institute (www.sas.com), Verizon (www.verizon.com), Container Store (www.containerstore.com), Google (www.google.com), Steelcase (www.steelcase.com), Whole Foods (www.wholefoods.com), and TELUS (www.telus.com). Choose one of these companies' websites and reviews of the companies at www.glassdoor.com and perform an organizational needs analysis. Read about the company's values and vision; look for statements about the importance of training and personal development. Is training important in the company? Why or why not? Provide supporting evidence.
5. Go to www.careeronestop.org/CompetencyModel, the website for CareerOneStop, sponsored by the U.S. Department of Labor, Employment and Training Administration to help job seekers, students, businesses, and career professionals. Choose and review one of the industry competency models. How might this competency model be useful for training and development for companies within the industry you selected? For their employees? For individuals such as students or the unemployed interested in working in the industry?
6. ConocoPhillips finds and produces oil and natural gas. Go to www.conocophillips.com. Click on "Careers." Under "Career Development," click on "Leadership Competencies." Review the competencies. How could these competencies be used in the learning and development of new and aspiring company leaders and managers?
7. O*NET is the primary source of occupational information in the United States. The O*NET database includes information on skills, abilities, knowledge, work activities, interests, and more. Go to www.onetonline.org. Click on "Find Occupations" and choose an occupation that interests you. Review the information provided for the occupation. How could this information be used for training needs assessment?
8. Go to www.designingdigitally.com, the website for Designing Digitally, a company that provides custom training solutions. Under "What We Do," click on "Training Needs Analysis." Read the information provided. What role does needs analysis have at Designing Digitally? How does needs assessment help Designing Digitally develop training that is best for its clients? What questions might Designing Digitally ask about the organization in its needs analysis?

Case: Identifying Training Needs at the U.S. Department of Agriculture's Animal Plant and Health Inspection Service (APHIS)

APHIS created an innovative training program to keep supervisors up to date and motivated. The Experienced Supervisor Applied Workshop directs managers to focus on the spirit and practice of effective management as well as current policy changes. The purpose of the program was to provide in-person training that would meet the organization's mandatory supervisory training requirement. The training focused on how to mentor employees, improve employee performance and productivity, conduct performance appraisals, and deal with unacceptable performance. To help ensure that this new training program met the organization's needs, the planning team conducted interviews with supervisors who had at least two years of management experience in the agency to determine what kind of training would be most helpful to them. The team also met with sub-

ject-matter experts to determine where they saw the biggest need for improvement. Also, the planning team analyzed participant feedback from an already existing training program for new supervisors.

In addition to the interviews, meetings with subject-matter experts, and analysis of participant feedback from an existing training program, what else could APHIS have done to conduct the needs assessment? Describe the method, why you would recommend it, and who would be involved. Would you recommend your method replace or be used in addition to the methods already in use? Explain.

Source: Based on B. Epstein and M. Schweitzer, "Not Your Ordinary Government Training," *T+D*, July 2000, pp. 56–61.

Endnotes

1. I. L. Goldstein, E.P. Braverman, and H. Goldstein, "Needs Assessment," in *Developing Human Resources*, ed. K. N. Wexley (Washington, DC: Bureau of National Affairs, 1991), pp. 5-35 to 5-75; E. Surface, "Training Need Assessment: Aligning Learning and Capability with Performance Requirements and Organizational Objectives," in *The Handbook of Work Analysis: Methods, Systems, Applications and Science of Work Measurement in Organizations*, eds. M.A. Wilson, W. Bennett, S. G. Gibson, and G. M. Alliger (Routledge Academic, 2012), pp. 437–462.
2. D. Mahoney, "You Still Need to Do the Work," *training*, March/April 2020, p. 108; C. Lacerenza, D. Reyes, S. Marlow, D. Joseph, and E. Salas, "Leadership Training Design, Delivery, and Implementation: A Meta-Analysis," *Journal of Applied Psychology* 102 (2017), pp. 1686–718; K. Tyler, "The Strongest Link," *HR Magazine*, January 2011, pp. 51–53; S. Villachica, D. Stepich, and S. Rist, "Surviving Troubled Times: Five Best Practices for Training Solutions," *Performance Improvement* 50 (March 2011), pp. 9–15.
3. A. Rio, "Better Together," *Chief Learning Officer*, June 2019, pp. 54–55; "Training 2021 Top 100 Rankings: ArcBest," *training*, March 2021, p. 49.
4. A. Rio, "Small, Yet Mighty," *Chief Learning Officer*, June 2019, pp. 42–43; S. Castellano, "Rewriting the Rules," *T+D*, October 2019, pp. 62–64.
5. J. Congemi, "MasTec Tackles the LMS," *training*, July/August 2014, pp. 52–54.
6. S. Castellano, "A Certified Success," *T+D*, October 2017, pp. 52–54.
7. J. Donahue, "Immersed in Leadership," *T+D*, January 2021, pp. 24–26.

8. M. Weinstein, "Partners in Learning," *training*, March/April 2018, pp. 22–25.
9. G. Siekierka, H. Huntley, and M. Johnson, "Learning is at Center stage at CSC," *T+D*, October 2009, pp. 48–50.
10. L. Freifeld, "LeadingRE's #1 Listing," *training*, January/February 2018, pp. 22–28.
11. L. Freifeld, "All Eyes on Safety Training," *training*, January/February 2018, p. 8; "What Is Eye Tracking?" from www.tobii.com website for tobii tech, accessed February 25, 2018.
12. K. Ellis, "The Right Track," *training*, September 2004, pp. 40–45.
13. M. Casey and D. Doverspike, "Training Needs Analysis for New Technologies Through the Use of Problem-Based Inquiry," *Performance Improvement Quarterly* 18 (1) (2005), pp. 110–24.
14. S. Castellano, "Preparing for the New World of Banking," *T+D*, 2020 Best Practices, pp. 19–20.
15. "ATD 2019 Best Award Winner: Samsung Electronics," *T+D*, October 2019, p. 47.
16. L. Day, "Benchmarking Training," *Training and Development*, November 1995, pp. 27–30.
17. Association for Talent Development, "ATD Forum: Connect.Collaborate.Learn," from atd-forum.td.org, accessed March 4, 2021.
18. M. Mihelich, "Change is a Way of Life, Learning at IBM," *Chief Learning Officer*, June 2014, pp. 28–29.
19. J. Rouiller and I. Goldstein, "The Relationship Between Organizational Transfer Climate and Positive Transfer of training," *Human Resource Development Quarterly* 4 (1993), pp. 377–90; R. Noe and J. Colquitt, "Planning for Training Impact," in *Creating, Implementing, and Managing Effective Training and Development*, ed. K. Kraiger (San Francisco: Jossey-Bass, 2002), pp. 53–79.
20. S. Sipek, "Learning Developed with Surgical Precision," *Chief Learning Officer*, June 2015, pp. 22–25.
21. G. Rummler, "In search of the holy performance grail," *Training and Development*, April 1996, pp. 26–31; D. Langdon, "Selecting interventions," *Performance Improvement* 36 (1997), pp. 11–15.
22. K. Gupta, *A Practical Guide to Needs Assessment* (San Francisco: Jossey-Bass/Pfeiffer, 1999).
23. K. Bauer, K. Orvis, K. Ely, and E. Surface, "Re-examination of Motivation in Learning Contexts: Meta-analytically Investigating the Role Type of Motivation Plays in the Prediction of Key Training Outcomes," *Journal of Business and Psychology* 31 (2016), pp. 33–50; R. A. Noe, "Trainee Attributes and Attitudes: Neglected Influences on Training Effectiveness," *Academy of Management Review* 11 (1986), pp. 736–49; R. Noe and J. Colquitt, "Planning for Training Impact," in *Creating, Implementing, and Managing Effective Training and Development*, ed. K. Kraiger (San Francisco: Jossey-Bass, 2002), pp. 53–79.
24. D. Milibank, "Marriott Tightens Job Program Screening," *The Wall Street Journal*, July 15, 1997, pp. A1, A12.
25. T. T. Baldwin, R. T. Magjuka, and B. T. Loher, "The Perils of Participation: Effects of Choice on Trainee Motivation and Learning," *Personnel Psychology* 44 (1991), pp. 51–66; S. I. Tannenbaum, J. E. Mathieu, E. Salas, and J. A. Cannon-Bowers, "Meeting Trainees' Expectations: The Influence of Training Fulfillment on the Development of Commitment, Self-Efficacy, and Motivation," *Journal of Applied Psychology* 76 (1991), pp. 759–69; J. Colquitt, J. LePine, and R. Noe, "Toward an Integrative Theory of Training Motivation: A Meta-Analytic Path Analysis of 20 Years of Research," *Journal of Applied Psychology* 85 (2000), pp. 678–707.
26. M. Eisenstein, "Test, then train," *T+D*, May 2005, pp. 26–27.
27. J. Nunally, *Psychometric Theory* (New York: McGraw-Hill, 1978).
28. L. Gottfredson, "The G Factor in Employment," *Journal of Vocational Behavior* 19 (1986), pp. 293–96.

29. M. J. Ree and J. A. Earles, "Predicting Training Success: Not Much More Than G," *Personnel Psychology* 44 (1991), pp. 321–32.
30. S. McCartney, "The Air-Traffic Cops go to School," *The Wall Street Journal*, March 29, 2005, pp. D1, D7.
31. D. R. Torrence and J. A. Torrence, "Training in the Face of Illiteracy," *Training and Development Journal*, August 1987, pp. 44–49.
32. National Immigration Forum, "Upskilling New Americans: Innovative English Training for Career Advancement," November 13, 2019, from immigrationforum.org, accessed February 25, 2021.
33. American Society for Training and Development, *Bridging the Skills Gap* (Alexandria, VA: American Society for Training and Development, 2009); R. Davenport, "Eliminate the Skills Gap," *T+D*, February 2006, pp. 26–31; R. Zamora, "Developing Your Team Members' Basic Skills," *Wenatchee Business Journal*, August 2005, p. C8; M. Davis, "Getting Workers Back to Basics," *Training and Development*, October 1997, pp. 14–15; J. House, "Factory Helps Teens Get Diplomas," *The Wall Street Journal*, August 8, 2014, pp. B1, B6.
34. D. Cope, "English Classes Help Retain Immigrant Workers," *Society for Human Resource Management*, February 20, 2021, from shrm.org/hr-today, accessed February 25, 2021.
35. D. Belkin, "More Companies Teach Workers What Colleges Don't," *The Wall Street Journal*, March 22, 2018, from www.wsj.com.
36. M. Edwards and J. Smola, "Career Advancement," *The Columbus Dispatch*, September 22, 2017, p. F19.
37. M. E. Gist, C. Schwoerer, and B. Rosen, "Effects of Alternative Training Methods on Self-Efficacy and Performance in Computer Software Training," *Journal of Applied Psychology* 74 (1990), pp. 884–91; J. Martocchio and J. Dulebohn, "Performance Feedback Effects in Training: The Role of Perceived Controllability," *Personnel Psychology* 47 (1994), pp. 357–73; J. Martocchio, "Ability Conceptions and Learning," *Journal of Applied Psychology* 79 (1994), pp. 819–25.
38. R. A. Noe and N. Schmitt, "The Influence of Trainee Attitudes on Training Effectiveness: Test of a Model," *Personnel Psychology* 39 (1986), pp. 497–523.
39. "Training 2021 Top 100 Rankings: ESL Credit Union," *training*, March 2021, p. 49.
40. M. A. Quinones, "Pretraining Context Effects: Training Assignments as Feedback," *Journal of Applied Psychology* 80 (1995), pp. 226–38; T. T. Baldwin, R. J. Magjuka, and B. T. Loher, "The Perils of Participation," *Personnel Psychology* 44 (1991), pp. 51–66.
41. P. Ackerman and R. Kanfer, "Work in the 21st Century: New Directions for Aging and Adult Development," *American Psychologist* 75 (2020), pp. 486–98; J. Kubeck, N. Delp, T. Haskett, and M. McDaniel, "Does Job-Related Training Performance Decline With Age?" *Psychology and Aging* 11 (1996), pp. 92–107; J. Bersin and T. Chamorro-Premuzic, "The Case for Hiring Older Workers," *Harvard Business Review*, September 26, 2019, from www.hbr.org, accessed February 11, 2021.
42. C. Houck, "Multigenerational and Virtual: How Do We Build a Mentoring Program for Today's Workforce?" *Performance Improvement* 50 (February 2011), pp. 25–20; K. Ball and G. Gotsill, *Surviving the Baby Boom Exodus* (Boston: Cengage Learning, 2011); R. Zemke, C. Raines, and B. Filipezak, "Generation Gaps in the Classroom," *Training*, November 2000, pp. 54–58; J. Salopek, "The Young and Rest of Us," *Training and Development*, February 2000, pp. 26–30; A. Ort, "Embrace Differences when Training Intergenerational Groups," *T+D*, April 2014, pp. 60–65.
43. L. H. Peters, E. J. O'Connor, and J. R. Eulberg, "Situational Constraints: Sources, Consequences, and Future Considerations," in *Research in Personnel and Human Resource Management*, eds. K. M. Rowland and G. R. Ferris (Greenwich, CT: JAI Press, 1985), pp. 79–114; E. J. O'Connor, L. H. Peters,

- A. Pooyan, J. Weekley, B. Frank, and B. Erenkranz, "Situational Constraints Effects on Performance, Affective Reactions, and Turnover: A Field Replication and Extension," *Journal of Applied Psychology* 69 (1984), pp. 663–72; D. J. Cohen, "What Motivates Trainees?" *Training and Development Journal*, November 1990, pp. 91–93; J. S. Russell, J. R. Terborg, and M. L. Power, "Organizational Performance and Organizational Level Training and Support," *Personnel Psychology* 38 (1985), pp. 849–63.
44. L. Freifeld, "Why Cash doesn't Motivate. . ." *training*, July/August 2011, pp. 16–20.
 45. W. D. Hicks and R. J. Klimoski, "Entry into Training Programs and its Effects on Training Outcomes: A Field Experiment," *Academy of Management Journal* 30 (1987), pp. 542–52.
 46. J. Murray, "Get to the Root of the Problem," *T+D*, February 2017, pp. 26–27, 29; R. F. Mager and P. Pipe, *Analyzing Performance Problems: Or You Really Oughta Wanna*, 2d ed. (Belmont, CA: Pittman Learning, 1984); A. P. Carnevale, L. J. Gainer, and A. S. Meltzer, *Workplace Basics Training Manual* (San Francisco: Jossey-Bass, 1990); G. Rummier, "In Search of the Holy Performance Grail," *Training and Development*, April 1996, pp. 26–31; C. Reinhart, "How to Leap over Barriers to Performance," *Training Development*, January 2000, pp. 46–49; R. Jaenke, "Identify the Real Reasons Behind Performance Gaps," *T+D*, August 2013, pp. 76–77; E. Holton, R. Bates, and S. S. Naquin, "Large-scale Performance-Driven Training Needs Assessment," *Public Personnel Management* 29 (2000), pp. 249–67.
 47. G. A. Rummier and K. Morrill, "The Results Chain," *T+D*, February 2005, pp. 27–35; D. LaFleur, K. Smalley, and J. Austin, "Improving Performance in a Nuclear Cardiology Department," *Performance Improvement Quarterly* 18(1) (2005), pp. 83–109.
 48. "ATD 2019 Best Award Winner: Papa Murphy's International," *T+D*, October 2019, p. 87.
 49. Z. Khan, "Coming Up Short," *T+D*, January 2020, pp. 54–59; C. Schneier, J. Guthrie, and J. Olian, "A Practical Approach to Conducting and Using Training Needs Assessment," *Public Personnel Management*, Summer 1988, pp. 191–205; J. Annett and N. Stanton, "Task Analysis," *International Review of Industrial and Organizational Psychology* 21 (2006), pp. 45–74.
 50. I. Goldstein, "Training in Organizations," In *Handbook of Industrial/Organizational Psychology*, 2nd ed., eds. M. D. Dunnette and L. M. Hough (Palo Alto, CA: Consulting Psychologists Press, 1991), pp. 507–619.
 51. Based on Insurance Company presentation to MHR 7314 Designing Learning Systems Class, The Ohio State University, September 2018.
 52. M. Campion, A. Fink, B. Ruggeberg, L. Carr, G. Phillips, and R. Odman, "Doing Competencies Well: Best Practices in Competency Modeling," *Personnel Psychology* 64 (2011), pp. 225–262; J. Shippman, R. Ash, M. Battista, L. Carr, L. Eyde, B. Hesketh, J. Kehow, K. Pearlman, and J. Sanchez, "The Practice of Competency Modeling," *Personnel Psychology* 53 (2000), pp. 703–740; A. Lucia and R. Lepsinger, *The Art and Science of Competency Models* (San Francisco, Jossey-Bass, 1999).
 53. J. Shippman, R. Ash, M. Battista, L. Carr, L. Eyde, B. Hesketh, J. Kehow, K. Pearlman, and J. Sanchez, "The Practice of Competency Modeling," *Personnel Psychology* 53 (2000), pp. 703–40.
 54. A. Lucia and R. Lepsinger, *The Art and Science of Competency Models* (San Francisco, Jossey-Bass, 1999).
 55. J. Kochanski, "Competency-Based Management," *Training and Development*, October 1997, pp. 41–44; D. Dubois and W. Rothwell, "Competency-Based or a Traditional Approach to Training," *T+D*, April 2004, pp. 46–57; E. Kahane, "Competency Management: Cracking the Code," *T+D*, May 2008, pp. 71–76.

56. F. Morgeson, K. Delaney-Klinger, M. Mayfield, P. Ferrara, and M. Campion, "Self-Presentation Processes in Job Analysis: A Field Experiment Investigation Inflation in Abilities, Tasks, and Competencies," *Journal of Applied Psychology* 89 (2004), pp. 674–86; F. Lievens and J. Sanchez, "Can Training Improve the Quality of Inferences Made by Raters in Competency Modeling? A Quasi-Experiment," *Journal of Applied Psychology* 92 (2007), pp. 812–19; F. Lievens, J. Sanchez, and W. DeCorte, "Easing the Inferential Leap in Competency Modeling: The Effects of Task-Related Information and Subject Matter Expertise," *Personnel Psychology* 57 (2004), pp. 881–904.
57. A. Lucia and R. Lepsinger, *The Art and Science of Competency Models* (San Francisco: Jossey-Bass, 1999); M. Derven, "Lessons Learned," *T+D*, December 2008, pp. 68–73; M. Campion, A. Fink, B. Ruggerberg, L. Carr, G. Phillips, and R. Odman, "Doing Competencies Well: Best Practices in Competency Modeling," *Personnel Psychology*, 64 (2011), pp. 225–62."
58. C. Umbrell, "A Remedy for Nursing Needs," *T+D*, 2020 Best Practices, pp. 43.
59. "Voya's Leadership Model," Voya, from www.voya.com/page/why-voya, accessed February 2, 2021.
60. K. Gupta, *A Practical Guide to Needs Assessment* (San Francisco: Jossey-Bass, 1999); R. Zemke, "How to Do a Needs Assessment when You Don't Have the Time," *training*, March 1998, pp. 38–44; G. Piskurich, *Rapid Instructional Design* (San Francisco, CA: Pfeiffer, 2006); R. Tartell, "Use Focus Groups for Rapid Needs Assessment," *training*, March/April 2014, p. 14.
61. B. Hall, "Learning Analytics at JetBlue," *Chief Learning Officer*, November 2014, p. 16.
62. "KLA-Tencor Corporation: Right People, Right Knowledge," *training*, January/February 2015, pp. 56–57; "About Us," KLA, from www.kla-tencor.com, accessed March 4, 2021.